Representing Biomedical Experiments

A conceptual framework, its partonomy and its description-logic axiomatization

with the challenge–rescue experiment as a case study

Technical report on the experiment layer of the robot science assistant on Cell 01 of the LabAutomationTwin

[Authors and affiliations]

Generated	2026-10-17, from the generators' outputs
Case study	challenge–rescue experiment, design `c_d85_t48_n1`
Ontology	174 concepts, 52 object and 12 data properties, 246 TBox axioms (OWL 2 DL), HermiT-checked
Status	research prototype; biology is a surrogate, quantities ILLUSTRATIVE (D016)

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1 Executive summary

A robot scientist has to do more with an experiment than execute it. It has to know why each step is there, which culture plays which part, what a measurement stands for, which comparison answers which question, and whether the design can support the conclusion it is meant to support—ideally before anything is pipetted. This report describes how the experiment layer of the robot science assistant (RSA) on Cell 01 of the LabAutomationTwin came to represent experiments in that way, and what the resulting representation can and cannot check.

The case. The running example is a *challenge–rescue* experiment on organoid cultures (Section 4): two cultures are injured with a challenge agent, one of them is given a rescue agent 48 h later, an untreated culture serves as control, and organoid size is imaged at 72 h. The experiment discriminates three recovery hypotheses—graded recovery, a threshold ("point of no return"), and a protective response that moves the threshold. The design `c_d85_t48_n1` was selected for its expected information gain of 0.5439 bit over the hypotheses. All biology is Cell 01's illustrative surrogate (D016).

From action lists to experiment logic. The representation was developed in nine steps (Section 5): from a culture-level plan of 12 actions ordered by causal links, through CRAM/RPL plans and entity designators, to the observation that the experiment's *logic* is three cultures, each defined by its treatment, compared pairwise so that each comparison isolates one exposure. Written that way (`experiment.ldl`), the experiment is short, has no variables, and the 13 culture actions and their order are *derived* from it; the derivation reproduces all 12 actions of the hand-written plan and adds the validity analysis that the hand-written plan only implied.

A conceptual framework. Generalising from the case, Section 7 proposes a framework of five layers—epistemic, design, protocol, execution and biological system—that keeps apart what is *planned*, what *happens*, what is *measured* and what is *believed*. Its core concepts (units, factors and levels, arms, controls defined by what they control for, replicates, readouts as proxies, comparisons, interaction contrasts, analysis plans, hypotheses that *posit* biology, belief and information gain, deviations, conclusions) organise a multi-axial taxonomy of experiments. Validity criteria are assigned to the mechanism that can enforce them: description logic for what experiments *are*, closed-world queries for what a design *must contain*, run-time monitors for what an execution *must keep*, and probabilistic models for what it *will tell us*. Three partonomies (process, material, plan) are separated from containment, membership, participation and realization (Section 8).

Axiomatization and reasoning. The framework is axiomatized in OWL 2 DL (Section 11): 246 TBox axioms over 174 concepts in 13 groups, 71 of them definitions, so that the HermiT reasoner *classifies* arms, comparisons, designs and experiments from their content. Closure axioms in the ABox make the definitions with negation decidable. The ontology is consistent alone, with both designs, and together with Cell 01's biology ontology; all 8 negative controls are detected as inconsistencies; 30 subsumptions between experiment classes are inferred rather than stated. What OWL 2 cannot express—single-factor contrasts, replicate counts, plate-edge placement, factorial coverage, interaction estimability—is checked by closed-world queries (Section 12).

The main finding. The reasoner classifies the current experiment as a *validated injury–rescue* experiment that is hypothesis-discriminating, sufficiently informative, pre-specified and well contrasted, but *not* as adequately controlled: it lacks a rescue-only arm (so the interaction of injury and rescue is not estimable), a vehicle arm, a positive control and replicates, and its wells lie on the plate edge. An improved, ILLUSTRATIVE design—a challenge × time-of-rescue factorial with 5 arms, 15 units and 5 comparisons including a therapeutic-window contrast—is classified as an *adequately controlled injury–rescue design*, and each variant that removes one ingredient loses exactly the corresponding class (Section 13). The critique a biomedical reviewer would make is thus available as a formal result before the experiment is run.

What our laboratory affords. Laboratories are built for classes of experiments, so the ontology also describes laboratories (Section 14). Capability classes are defined by the functions a laboratory's devices bear, its environments and its validated readouts; requirement classes are defined by a design's content. HermiT classifies Cell 01, from its 10 component records, as a treatment-and-imaging module that is dosing, cultivation, imaging, quality-controlled, containment-capable and connected, but not an endpoint-reader or multi-environment laboratory. Matched against it, the current design is executable in Cell 01; the improved design lacks only a vehicle stock and, with it, needs two dosing sessions per time point (at most 11 wells fit an excursion); an ATP endpoint is executable on site with a transfer to an ILLUSTRATIVE core plate reader; and a design with an oxygen factor is not executable, because no laboratory provides two environments.

Limits. The CRAM plans are not executed (no Lisp CRAM is installed); the factors and analysis plan exist in the OWL layer but not yet in the authored LDL; BFO and OBI are followed in style, not imported; the replication minimum and the information target are illustrative criteria; in vivo and clinical experiments are outside the current taxonomy (Section 17).

2 Motivation and scope

Laboratory automation has made it routine for robots to *perform* protocols. Robot scientists such as Adam and Eve [1, 2] went further and closed the loop from hypothesis to experiment to analysis, and self-driving laboratories in chemistry and materials science [3, 4] have shown how far closed-loop experimentation can go when the experiment space is well parameterised. Biomedical experiments are harder to parameterise: the system is alive, the quantities of interest are hidden and reached only through proxies, and whether an experiment supports its conclusion depends on its *logic*—which cultures are compared, what they differ in, which controls exclude which alternative explanation—more than on any single step. Poor design is one of the recognised causes of irreproducible biomedical results [5], and reporting guidelines such as ARRIVE [6] and CONSORT [7] exist largely to make that logic explicit after the fact.

For a robot scientist the logic has to be explicit *before* the fact, in a form a program can inspect. The RSA already derives culture-level plans, refines them into lab actions and motions, and audits their execution (see the companion report on the RSA kernel). What it lacked was a representation of the experiment *as an experiment*: of its question and hypotheses, its units and their roles, its treatments, its comparisons and their purpose, and the standards against which a design is judged. This report fills that gap. Its contributions are:

1. a sequence of representations of one experiment, from an ordered list of actions to a variable-free entity designator organised by the experiment's logic, with the actions and their order derived and checked (Sections 4–6);
2. a conceptual framework for biomedical experiments in five layers, with core concepts, a multi-axial taxonomy, validity criteria assigned to enforcement mechanisms, and a lifecycle (Section 7);
3. a partonomy that separates process, material and plan parthood from containment, membership, participation and realization (Section 8);
4. two formalisations: an LDL concept hierarchy with a closed-world checker (Section 9) and an OWL 2 DL axiomatization with defined concepts, classified and checked by HermiT (Sections 10–13);
5. an evaluation on the case study and on an improved design, with negative controls at every level.

The scope is the in-vitro injury–rescue family of experiments; the framework is written to extend beyond it, and Section 17 lists what that extension still requires.

3 Background

3.1 Experimental design

The vocabulary of controlled experimentation goes back to Fisher [8]: experimental units, treatments, replication to estimate chance variation, randomisation to protect against systematic bias, and blocking to remove known sources of variation. For laboratory biology, Lazic [9] gives a modern account; it stresses that the experimental unit is the entity *independently* assigned to a treatment (a culture, not a measurement of it), that technical and biological replicates answer different questions, and that controls are defined by what they rule out. Factorial designs vary several factors at once and make their interactions estimable. Clinical trials use the same notions under other names: arms, epochs and the cells of an arm-by-epoch grid, as formalised in the CDISC Study Design Model [10].

3.2 Ontologies of experiments

Ontologies of scientific experiments were proposed early: EXPO [11] describes experiments as investigations with hypotheses, designs, objects and results; EXACT [12] describes protocols as sequences of actions. The Ontology for Biomedical Investigations (OBI) [13, 14], part of the OBO Foundry [15], models investigations as planned processes that realize plans, assays with specified inputs and outputs, study designs and their independent variables, and roles borne by material entities; it is built on the Basic Formal Ontology (BFO) [16], which separates continuants (material entities, qualities, roles, dispositions) from occurrents (processes). The ISA framework [17] carries investigation, study and assay metadata for data deposition. The framework of this report follows BFO and OBI in style—planned processes execute plans, roles and dispositions are borne, data are about qualities—without importing them (Section 17).

Table 1: The three recovery hypotheses and their predicted rescue effects for design `c_d85_t48_n1` (surrogate, ILLUSTRATIVE).

hypothesis	claim	predicted effect (µm)
M1 graded	recovery declines smoothly with injury and delay; no point of no return	8.5 ± 11.4
M2 threshold	once cumulative injury exceeds a threshold, the culture loses its capacity to recover	25.0 ± 6.3
M3 protective	as M2, but the adaptive response raises the threshold	25.4 ± 6.5

3.3 Description logics and OWL

Description logics [18] are decidable fragments of first-order logic built from concepts and roles. $\mathcal{SROIQ}(\mathbf{D})$ [19] underlies OWL 2 DL [20]: it adds role hierarchies, transitivity, inverse roles, role chains, qualified number restrictions and datatypes. Number restrictions are allowed only on *simple* roles (roles that are neither transitive nor composite, and have no such sub-roles). OWL reasons under the open-world assumption: what is not stated is unknown, not false. Checking that a design *contains* something is therefore a closed-world question, usually answered with constraint languages such as SHACL [21] or with integrity-constraint semantics [22]. Concrete domains with feature chains [23] can compare data values reached by different paths, which OWL 2 cannot. HermiT [24] is a hypertableau reasoner for OWL 2; Owlready2 [25] embeds ontologies and reasoners in Python.

3.4 Plans and Bayesian design in the RSA

The RSA represents plans in a lab designator language (LDL) and executes them as structured reactive controllers in the style of RPL and CRAM [26–28]; knowledge comes from a KnowRob-style ontology [29]. Designs are chosen by expected information gain [30, 31]: the mutual information between the hypotheses and the outcome of a candidate design, estimated by nested Monte Carlo over Cell 01's surrogate.

4 The challenge–rescue experiment

4.1 System, question and hypotheses

The units are organoid cultures in wells of a microplate, kept in an incubator under a standard culture environment. Cell 01's surrogate describes each culture by hidden states—injury, an adaptive protective response, a glutathione-like protective reserve, and viability—driven by two abstract agents: a *challenge* that raises injury and consumes the reserve, and a *rescue* that restores the reserve and supports recovery. The structure resembles classic toxicology paradigms in which an antidote's efficacy depends on how late it is given; the project names no agent, doses are in arbitrary units (au), and every coefficient is a placeholder (D016). Organoid cultures [32] are a natural target for such experiments because their morphology can be imaged without destroying them.

Earlier (simulated) data showed that a rescue at 6 h is effective at every challenge dose tested. The open question is what happens when the rescue comes much later. Table 1 lists the competing hypotheses and the effect each predicts for the selected design; the effect is the difference in organoid radius between the rescued and the challenge-only culture at 72 h.

4.2 The selected design

The planner scored 45 candidate designs (three challenge doses, five rescue delays, one to three wells per arm) by expected information gain and selected the cheapest design that reaches the required 0.5 bit (Figure 1). Expected gain grows with the delay, because the hypotheses agree at short delays (all were calibrated to the effective 6 h rescue) and diverge only at long ones; at 48 h and the highest dose, one well per arm already gives 0.5439 bit, and two or three wells give 0.601 and 0.641 bit. The resulting design has three cultures on plate P17: *rescued* (challenge 0.85 au at 0 h, rescue 0.70 au at 48 h, well A01), *comparator* (challenge only, A02) and *control* (untreated, A05), all imaged at 72 h.

4.3 What a run shows

Executed on the surrogate (Figure 2), the challenge shrinks both challenged cultures; after the rescue at 48 h the rescued culture partly recovers. At 72 h the radii are 201.6 µm (rescued), 186.8 µm (comparator) and 267.6 µm (control). The validity comparison shows that the challenge injured the cultures (-80.8 µm); the primary comparison gives a rescue effect of 14.8 µm. With a measurement noise of $\sigma \approx 2.5$ µm, the posterior over the hypotheses moves from uniform to M1 0.68, M2 0.15, M3 0.17: the entropy falls from 1.585 to 1.228 bit, a gain of 0.357 bit in this run against 0.5439 bit expected. In biological terms: no hard point of no return at 48 h, but a late rescue is clearly weaker than an

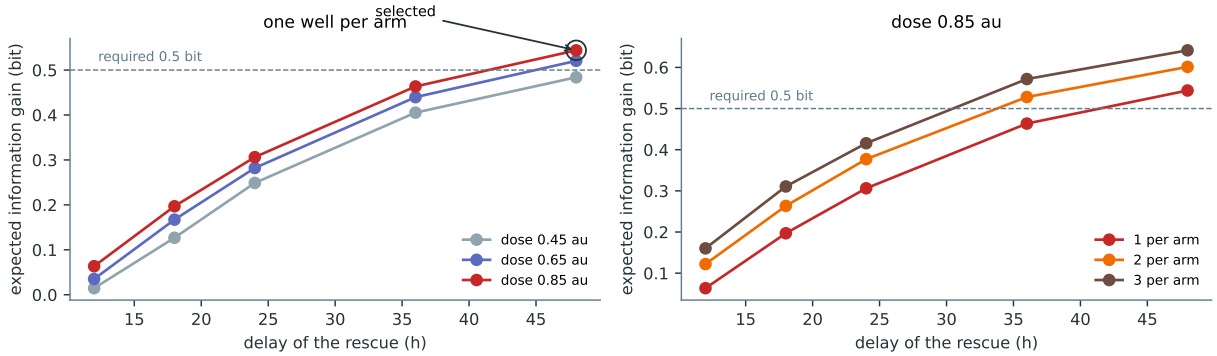


Figure 1: Expected information gain of the candidate designs over the three hypotheses. Left: one well per arm, by dose and delay; the selected design is circled. Right: the selected dose with one to three wells per arm. Surrogate, ILLUSTRATIVE.

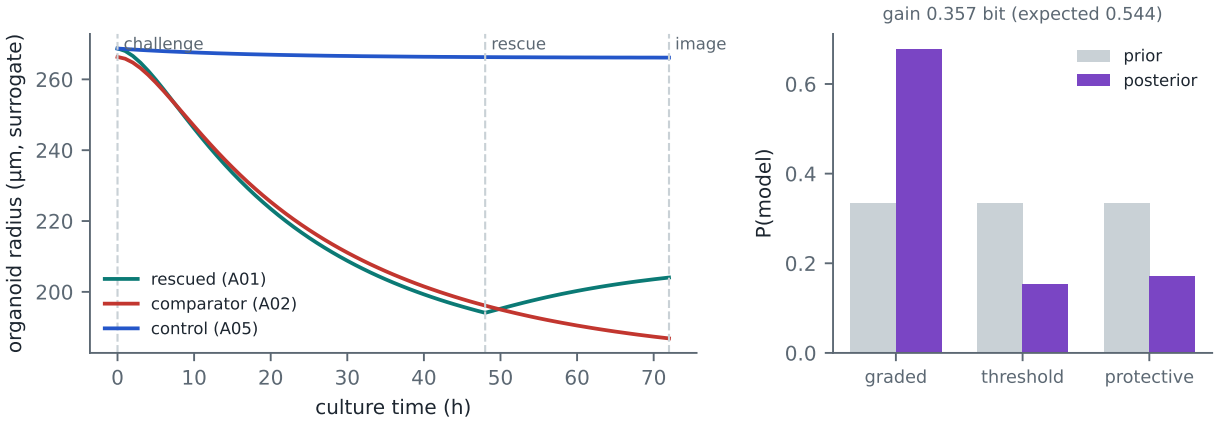


Figure 2: Surrogate run of the selected design: organoid radius of the three cultures over culture time (left) and the belief over the hypotheses before and after the analysis (right). ILLUSTRATIVE.

early one. M2 and M3 predict nearly the same effect in this design; it separates graded from threshold-like recovery, not the two threshold variants from each other.

4.4 How a biomedical reviewer would judge it

As a real experiment the design would not stand. It has one culture per arm, so culture-to-culture variation cannot be separated from the treatment effect. It has no rescue-only arm, so part of the rescue effect might be a general growth effect of the rescue agent rather than rescue of injury. Its control receives nothing while the treated wells receive liquid, so solvent and volume effects are confounded with the challenge. Its wells lie on the plate's edge row, where evaporation and temperature effects are strongest. It has a single delay and a single readout, and organoid size is only a proxy for viability. These are exactly the kinds of defects that a representation of the experiment *as an experiment* should make visible; Section 13 shows that the final axiomatization does.

5 From action lists to experiment logic

Table 2 summarises the representations the experiment went through. Each step was prompted by something the previous one made hard to see; each is generated from the one before it or from its source, and each was checked mechanically.

5.1 The culture-level plan

The first representation states only what is done to the cultures, how they are observed and how the observations are interpreted and analysed; no plate, instrument or robot appears. Every action states a *physical* effect (on the cultures' state) and a *knowledge* effect (records, images, estimates, the belief). The order is not written: it is derived from causal links, a link holding when an effect of one action entails a requirement of another. Listing 1 shows three

Table 2: Nine representations of the challenge–rescue experiment.

	representation	file	explicit	derived / checked
1	culture-level plan	`culture_plan.ldl`	12 actions with physical and knowledge effects	order from causal links (13: 7 physical, 6 knowledge)
2	CRAM/RPL controller	`culture_plan_src.lisp`	control flow, policies, failure handling	partial order = derived order; refinement into lab designators 8/8
3	experiment as an entity	`challenge_rescue_experiment.lisp`	question, materials, rules, steps, insights	one-line top-level plan: *conduct* the experiment
4	nested designators with roles	(superseded)	every participant fills a role; no variables	references identify one step each; after-roles = derived order
5	organised by logic	(superseded)	cultures with treatments, shared conditions, comparisons	single difference per comparison; 12/12 actions regenerated
6	authored experiment designator	`experiment.ldl`	the logic; validity comparison explicit	13 actions and order derived; vs. the hand-written plan
7	LDL concept hierarchy	`experiment_concepts.ldl`	5 experiment concepts, partonomy, procedure	closed-world conformance, classification, 8 negative controls
8	DL statement (v1)	`experiment_dl.pdf`	the LDL concepts in $\mathcal{SROIQ}(\mathbf{D})$	simple-role restriction; expressivity of each check
9	OWL 2 DL axiomatization (v2)	`experiment.ofn`	BFO-style biomedical ontology, defined classes	HermiT consistency, classification, realization; closed-world queries

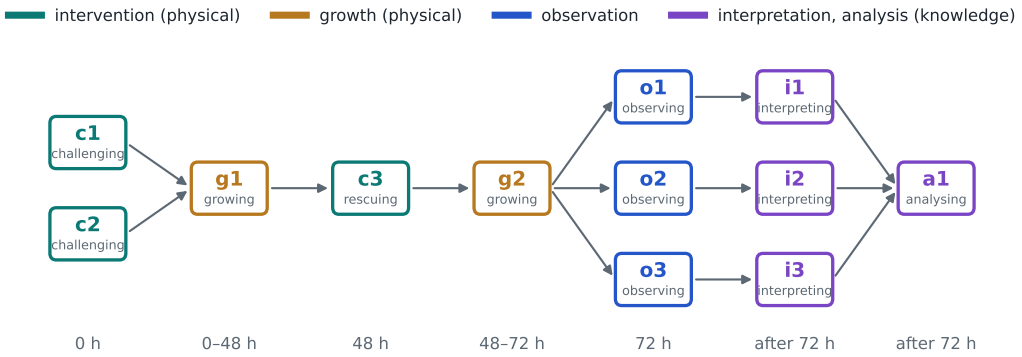


Figure 3: Order of the twelve culture actions, derived from causal links over physical (solid) and knowledge (dashed) effects.

of the 12 actions; Figure 3 shows the derived order. The plan is precise, but its logic is implicit: that the comparator exists to isolate the rescue, or that the control licenses the comparison, can only be reconstructed from requirement lists.

```
(an action ?c1 (type challenging) (culture ?cr) (substance ?ch) (dose 0.85 au) (at (culture-time 0 h))
  (physical (the culture ?cr (exposed-to ?ch 0.85 au)) (the culture ?cr (injury rising)))
  (knowledge (a record (of ?c1) (states ?cr ?ch 0.85 au 0 h))))
...
(an action ?a1 (type analysing) (belief ?B) (statistic (difference (radius ?cr) (radius ?cc)))
  (requires (an estimate (of ?cr) (organoid-radius))
            (an estimate (of ?cc) (organoid-radius))
            (an estimate (of ?ck) (organoid-radius)))                ; the control licenses the comparison
  (knowledge (the belief ?B (posterior updated)) (the belief ?B (information-gain measured)))))
```

Listing 1: Two of the twelve actions of the culture-level plan (`culture_plan.ldl`).

5.2 The plan as a CRAM/RPL controller

The culture plan was then generated as a top-level CRAM/RPL controller: one action plan per culture action type, each waiting for its requirements, performing its refinement (a lab action designator, or a computation for interpreting and analysing) and asserting its effects; a partial order of the twelve designators; and a top-level function that binds the cultures, substances and belief, verifies the given state, runs the plan under the plan's own policies (the control stays untreated, no contamination, standard environment) and restores the safe state on any exit. The generator checks that the partial order equals the derived order and that the refinement templates reproduce the lab designators of the lab-level plan for all eight lab-refined actions. The controller was correct but, as a representation of the experiment, opaque: the experiment's purpose was dispersed over control structures.

5.3 The experiment as an entity, and roles

The third and fourth representations turned the experiment into one entity designator—question, hypotheses, materials, rules, steps and insights—performed by a one-line top-level plan, `(perform (an action (type conducting) (experiment …)))`, whose generic action plan derives everything from the entity. Nesting entity designators with roles removed all variables: every participant fills a named role, and every later mention is a definite description that identifies one entity, such as `(the culture (role rescued))`; steps refer to the steps they follow by the shortest identifying description. The generator checks that every reference identifies exactly one step and that the `after` roles equal the derived order.

5.4 Organised by the experiment's logic

Organising the entity by steps still hid the logic. The experiment is three cultures, each defined by its treatment, kept under shared conditions and observed the same way; comparisons between cultures that differ in exactly one exposure determine the effects. Rescued and comparator differ only in the rescue, so their difference measures the rescue's effect; comparator and control differ only in the challenge, so their difference shows that the challenge injured the cultures—the precondition for there being anything to rescue. Written this way, the twelve actions follow from the structure and were regenerated exactly, and the second comparison, which the culture plan expressed only as a requirement of the analysis, became explicit.

6 The authored experiment designator

The organised form became the authored source, `experiment.ldl` (Listing 2). It has no variables and no lab entities. The generator `experiment_designator.py` checks that every comparison isolates exactly one exposure—the one whose effect it determines—and refuses to proceed otherwise; derives the culture actions (an intervention per exposure, growth between the treatment times, an observation and an interpretation per culture, an analysis per comparison, with a licensed comparison's analysis requiring the licensing one's finding); writes them in the format of the culture plan; and derives their order with the same causal-link machinery.

```
(an experiment
  (instance-of challenge-rescue-experiment)       ; registry/lab/experiment_concepts.ldl; checked by rsa/experiment_concepts.py
  (name challenge-rescue)
  (question (a question
              (asks (which (a recovery-model)))
              (explains (the rescue-efficacy (after (a delay (of 48 h)))))
              (candidate (a recovery-model (name M1_graded)))
              (candidate (a recovery-model (name M2_threshold)))
              (candidate (a recovery-model (name M3_protective)))))
  ;; the design guarantees the gain in expectation (e2m); what one execution yields is measured, not promised
  (goal (the belief (expected-information-gain (>= 0.5 bit))))
  (belief (a belief (over (M1_graded M2_threshold M3_protective)) (prior uniform) (entropy 1.585 bit)))

  ;; what the treatments are made of, and what exposure to them does to a culture
  (substance (a substance (role challenge) (causes (injury rising))))
  (substance (a substance (role rescue) (causes (recovery possible))))

  ;; the cultures: each is defined by its treatment
  (culture (a culture (role rescued) (state baseline)
            (treatment (a treatment
                         (exposure (an exposure (to (the substance (role challenge))) (dose 0.85 au) (at (culture-time 0 h))))
                         (exposure (an exposure (to (the substance (role rescue))) (dose 0.70 au) (at (culture-time 48 h))))))))
  (culture (a culture (role comparator) (state baseline)
            (treatment (a treatment
                         (exposure (an exposure (to (the substance (role challenge))) (dose 0.85 au) (at (culture-time 0 h))))))))
  (culture (a culture (role control) (state baseline)
            (treatment (no treatment))))

  ;; everything but the treatment is shared, so that a difference between cultures is a difference of treatment
  (shared-conditions (a condition
                       (environment culture-standard)
                       (from (culture-time 0 h))
                       (to (culture-time 72 h))
                       (observation (an observation
                                      (readout morphology)
                                      (at (culture-time 72 h))
                                      (side-effect (light-dose increased))      ; imaging is not free (Cell 01 L17)
                                      (interpreted-as (an estimate (quantity organoid-radius) (sigma 2.5 um)))
                                      (needs (a scale (of (an image)) (known true)))))))

  ;; the comparisons: each between two cultures whose treatments differ in exactly one exposure (checked)
  (comparison (a comparison (role validity)
                (between (the culture (role comparator)) (the culture (role control)))
                (determines (the effect (of (the substance (role challenge)))))
                (established-when (the effect (below 0 um) (by-more-than 2 sd)))   ; ILLUSTRATIVE criterion
                (licenses (the comparison (role primary)))))
  (comparison (a comparison (role primary)
                (between (the culture (role rescued)) (the culture (role comparator)))
                (determines (the effect (of (the substance (role rescue)))))
                (updates (the belief))))

  (prevents
    (an exposure (of (the culture (role control))) (to (some substance)))   ; the control stays untreated
    (an event (type contamination) (of (a culture))))
  (maintains
    (each culture (environment culture-standard)))

  ;; what the experiment produces
  (insight (an insight (role answer) (answers (the question)) (form (a posterior (over (the belief)))))
            (from (the comparison (role primary)))))
```

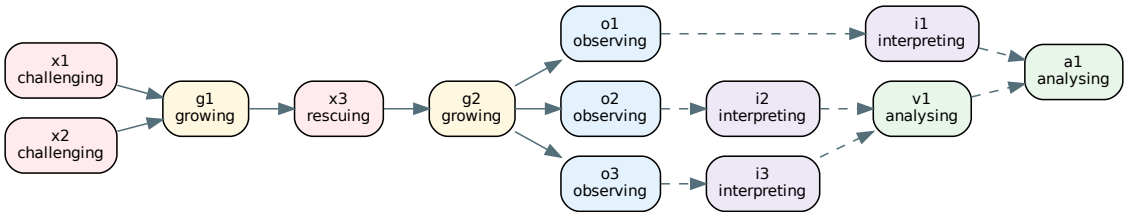


Figure 4: The 13 culture actions derived from `experiment.ldl`: interventions (red), growth (yellow), observations (blue), interpretations (violet), analyses (green); solid edges are physical links, dashed edges knowledge links.

```
(insight (an insight (role effect) (of (the substance (role rescue))) (from (the comparison (role primary)))))
(insight (an insight (role effect) (of (the substance (role challenge))) (from (the comparison (role validity))))))
```

Listing 2: The authored experiment designator (`rsa/registry/lab/experiment.ldl`).

The derivation yields 13 actions ordered by 15 causal links (7 physical, 8 knowledge): {x1, x2} < g1 < x3 < g2 < {o1, o2, o3} < {i1, i2, i3} < v1 < a1 (Figure 4). Compared with the hand-written plan, all 12 of its actions are reproduced and one is added, the validity analysis `v1`; the primary analysis now waits for v1's finding that the challenge's effect is established, so the belief is not updated unless the challenge worked. The order changes accordingly: the edges i2→a1, i3→a1 are replaced by i2→v1, i3→v1, v1→a1. Two effects that the hand-written plan states are not stated in the authored source ((the culture rescued (injury-burden accumulated)); (the culture rescued (viability settled))); neither affects the order.

The top-level plan performing the designator is one `perform` of a `conducting` action whose `experiment` role is filled by the entire designator, enriched with the placement chosen by the mapping to Cell 01 (plate P17, wells A01, A02, A05), the models' predictions and the checked single differences (Listing 3).

```
(def-top-level-cram-function challenge-rescue ()
  (perform (an action
               (type conducting)
               (experiment (an experiment
                               (instance-of challenge-rescue-experiment)
                               (name challenge-rescue)
                               (question (a question
                               ...
```

Listing 3: The head of the top-level plan (`rsa/out/challenge_rescue_designator.lisp`).

7 A conceptual framework for biomedical experiments

7.1 Core idea

An experiment is a planned process that intervenes on and observes a biological system, executing a design, in order to change a belief about hypotheses. The framework keeps four things apart because most errors in practice come from confusing them: what is *planned* (design and protocol), what *happens* (execution), what is *measured* (data) and what is *believed* (hypotheses, belief, conclusions). The biology is the hidden state in between, reached only through interventions and through readouts that are proxies for it. No biological fact or parameter is asserted by the framework: hypotheses *posit* biology, data are *proxies* (D016).

7.2 Five layers

The **epistemic layer** holds the question, the competing hypotheses, their predictions, the belief over them, the objective (here: discriminate the hypotheses with at least a target information gain) and the conclusions. The **design layer** holds the units, factors and levels, arms, controls, replication, allocation, readouts, comparisons and the analysis plan—everything decided before anything is done to a culture. The **protocol layer** details each kind of step (how to dose, incubate, image and analyse), device-independent and with tolerances. The **execution layer** is the run: phases, steps, participants, times, deviations, the refinement into lab actions and the audit trail. The

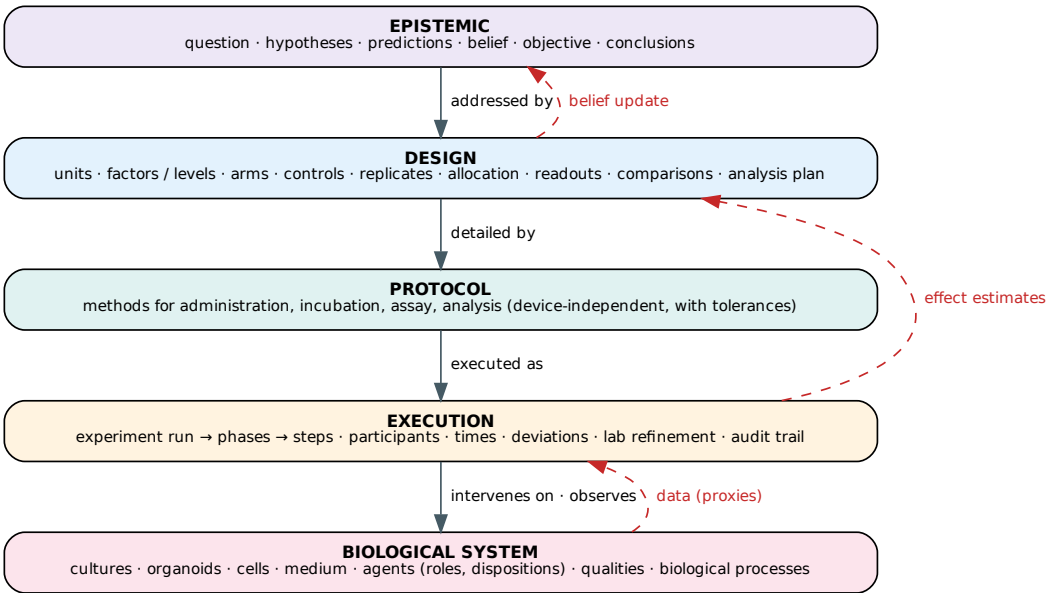


Figure 5: The five layers of the framework. Downward relations (black) connect the plan to the system; upward relations (red) carry evidence back to the belief.

Table 3: Core concepts of the framework.

concept	meaning
experimental unit	what is treated independently and measured: a culture in a well
factor, level	what is varied (agent, dose, time of rescue) and the values it takes
treatment	one combination of levels, given as a schedule of administrations
arm	a treatment and the units assigned to it; units are members of an arm, not its parts
control	an arm defined by what it controls for: negative (untreated, vehicle), positive (expected effect from prior evidence), agent-alone (the agent's own effect), comparator (isolates one factor)
replicate	units under the same treatment; biological (independent units) vs. technical (repeated measurement)
allocation	assignment and placement of units: randomisation, blocking, avoiding edge wells, blinding
readout	what is measured, when, by which assay, and what it is a proxy for (size for viability)
comparison	a pair of arms whose difference estimates the effect of exactly one factor
interaction contrast	a difference of comparisons: whether one factor's effect depends on another
validity check	a comparison that licenses another (the injury happened; the agent works)
analysis plan	one decision rule per comparison and a rule for deviations, fixed before the run
hypothesis, prediction	a model that posits processes, and its expected effect for a comparison
belief, information gain	a probability over hypotheses, and how much a design is expected to change it
deviation	a difference between execution and plan; it can invalidate comparisons
conclusion (insight)	a statement derived from effect estimates and the posterior, with provenance

biological system is the cultures, organoids, cells, medium, agents with their roles and dispositions, their qualities, and the biological processes they undergo. Each layer is linked to the next by one kind of relation (Figure 5): the experiment *executes* the design; steps *realize* parts of the plan; cultures *participate* in steps; data are *about* qualities and are *proxies for* hidden states; effect estimates *update* the belief; conclusions are *derived from* evidence.

7.3 A multi-axial taxonomy

Biomedical experiments are classified along independent axes, so the taxonomy is a polyhierarchy rather than a tree: the *system* (in silico, in vitro—cell line, primary cells, organoid—ex vivo, in vivo, clinical); the *purpose* (exploratory, confirmatory, hypothesis-discriminating, dose–response, time-course, screening); the *design structure* (comparative, controlled, factorial, within- or between-unit, randomised, blinded); the *intervention* (challenge, therapy or rescue, both); and *adequacy* (validated, agent-controlled, positive-controlled, replicated, pre-specified). Because classes are

Table 4: Validity criteria and the mechanism that enforces each.

validity	criteria	enforced by
internal	single factor per comparison; negative, agent-alone and positive controls; randomisation; blinding	defined classes (reasoner); closed-world checks
statistical	biological replicates; expected information or power	closed-world counts; information-gain model
construct	readout states its proxy relation; manipulation checked by a validity comparison	ontology; licensing order in the plan
executional	identity, contamination and environment kept; plan realized; deviations recorded	run-time monitors (CRAM policies); audit trail
external	relevance of the system to the target biology	stated as an assumption, never inferred

defined rather than merely named, a reasoner places an experiment under every class it satisfies; the challenge–rescue experiment is, at once, in vitro, hypothesis-discriminating, a controlled injury–rescue experiment and validated—but not adequately controlled.

7.4 Criteria and their enforcement

Table 4 assigns each kind of validity to the mechanism able to enforce it. The principle is a division of labour: *description logic for what experiments are; closed-world queries for what a design must contain; run-time monitors for what an execution must keep; probabilistic models for what it will tell us.* A design's adequacy is thus checked before the run, an execution's integrity during it, and the information it yields after it, each by the formalism suited to the question.

7.5 Lifecycle

The framework is used along a lifecycle: question → hypotheses and predictions → candidate designs → selection by information and cost → **adequacy check before the run** → protocol → refinement into lab actions → monitored execution → data → analysis by the pre-specified rules → belief update → conclusions → next question. Each step reads and writes one layer, and the audit trail links them. In the RSA, selection is the information-gain planner, the adequacy check is the reasoner of Section 11 together with the closed-world queries of Section 12, refinement and execution are the LDL and CRAM layers, and the belief update is the posterior over the recovery models.

7.6 Design principles

Five principles recur. *Plans are information.* A design is an information entity that a process executes; the same design can be run many times, and each run is classified by its design. *Membership is not parthood.* A culture is assigned to an arm; it is not part of it, and a level is shared by the arms at that level. *Roles and dispositions are borne.* An agent is a challenge because it bears that role in this experiment, and a toxicant because it bears a toxicity disposition; the role is given because of the disposition. *Data are proxies.* A size datum is about organoid size and is a proxy for viability, never viability itself. *Hypotheses posit.* A threshold hypothesis posits a loss of recovery capacity; nothing in the ontology asserts that it occurs.

8 Partonomy

Parthood is the backbone of any representation of experiments, and also its most common source of error: a culture is not part of an arm, a well is not part of a treatment, an instrument is not part of an assay. The framework therefore has one transitive parthood relation with three kinds of sub-relations, and keeps every other whole–part-like relation separate (Table 5). Parthood is *kind-preserving*: processes have processes as parts, material entities material entities, information entities information entities. The kinds of parthood are *simple* sub-roles of the transitive *hasPart*, so they may be counted (exactly three phases, at most four arms) without violating the OWL 2 restriction on number restrictions.

Figure 6 shows the three partonomies of the case study side by side. The process partonomy has the experiment, its three phases and the 13 steps; the material partonomy has the plate and its three wells, which *contain* (not have as parts) the three cultures; the plan partonomy has the treatments and their exposures. Interventions *realize* exposures—the dotted edges—which is how the execution is tied to the plan without either being part of the other.

Table 5: Parthood and the relations kept apart from it.

relation	kind	connects	example
hasPart	parthood	transitive; kind-preserving	—
hasPhase, hasStep	process parthood	experiment → phases → steps	cultivation phase has step x1
hasWell, hasOrganoid, hasCell, hasMedium	material parthood	plate → wells; culture → organoids, medium	P17 has well A01
hasArm, hasFactor, hasLevel, hasAdministrationSpec, hasComparisonSpec, hasReadout, hasAnalysisPlan, hasDecisionRule	plan parthood	design → arms, factors, comparisons, readout, analysis plan; arm → administrations	design has arm rescued
contains / containedIn	location	well contains culture; chain containedIn ∘ partOf ⊑ locatedIn	A01 contains the rescued culture
assignedTo	membership	unit → arm (functional)	rescued culture assigned to arm rescued
atLevel	plan reference	arm → factor level (shared between arms)	arm rescued at level rescue 0.7au_48h
hasParticipant	participation	chain hasPart ∘ hasParticipant ⊑ hasParticipant	x1 has input the rescued culture
executes	realization	process → plan part	x1 executes the challenge administration
bearerOf / inheresIn	dependence	continuant → role, disposition, quality	agent bears the challenge role

9 The LDL concept layer

Before the DL axiomatization, the experiment concepts were written in LDL itself (`experiment_concepts.ldl`): a specialization hierarchy of 33 concepts with 27 roles, each concept stating the roles its instances must fill (a pattern and a cardinality), the roles whose fillers are restricted, and named checks. Five concepts form the experiment hierarchy: *experiment*, then *comparative* and *hypothesis-discriminating*, then *controlled*, then *challenge–rescue*, which specializes both branches (Figure 7). The procedure that derives the actions is attached to *comparative-experiment* and inherited by every specialization. Listing 4 shows two concepts.

```
(a concept controlled-experiment
  (specializes comparative-experiment)
  (documentation "one unit is an untreated control, it stays untreated, and every comparison isolates exactly one
exposure: its two units differ in that exposure only, and it is the exposure whose effect the comparison determines")
  (has unit (an entity (role control) (treatment (no treatment))) (exactly 1))
  (has prevents (an exposure (of (the culture (role control)))) (at-least 1))
  (check single-difference))

(a concept challenge-rescue-experiment
  (specializes controlled-experiment hypothesis-discriminating-experiment)
  (documentation "units are injured by a challenge; one is then given a rescue after a delay. A validity comparison
(comparator vs control) establishes the challenge's effect and licenses the primary comparison (rescued vs
comparator), which determines the rescue's effect and updates the belief over models of recovery")
  (has substance (a substance (role challenge)) (exactly 1))
  (has substance (a substance (role rescue)) (exactly 1))
  (has unit (a culture (role rescued)
              (treatment (a treatment (exposure (an exposure (to (the substance (role challenge)))))
                                      (exposure (an exposure (to (the substance (role rescue))))))))
       (exactly 1))
  (has unit (a culture (role comparator) (treatment (a treatment (exposure (an exposure (to (the substance (role
  ↪  challenge))))))))
       (exactly 1))
  (has comparison (a comparison (role validity) (between (the culture (role comparator)) (the culture (role control)))
                    (determines (the effect (of (the substance (role challenge))))) (licenses (the comparison (role primary))))
       (exactly 1))
  (has comparison (a comparison (role primary) (between (the culture (role rescued)) (the culture (role comparator)))
                    (determines (the effect (of (the substance (role rescue))))) (updates (the belief)))
       (exactly 1))
  (check rescue-after-challenge))
```

Listing 4: Two concepts of the LDL concept layer (`experiment_concepts.ldl`).

The checker `experiment_concepts.py` verifies the experiment *as derived and placed*—with its phases and steps from the procedure and its plate and wells from the mapping—against its concept and all ancestors, checks every part against its own concept (50/50 parts conform), and classifies it independently. The declared concept is confirmed as the most specific one; all 36 constraints hold (Table 6). Each of the 8 negative controls—modified copies of the design or of its derived plan—fails exactly at the level whose constraint it breaks and is reclassified under the more general concepts it still satisfies (Table 7).

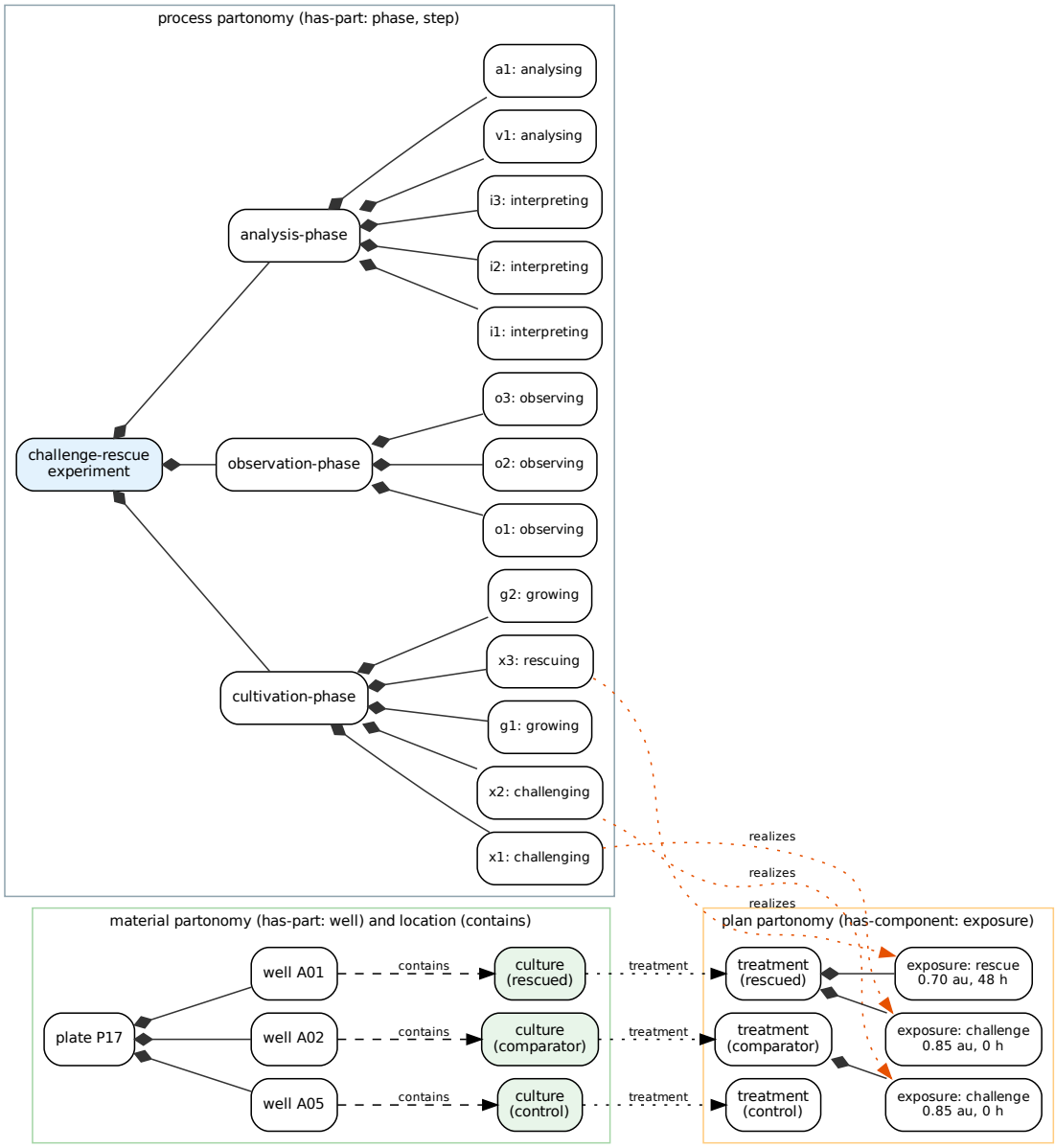


Figure 6: The three partonomies of the case study: process (top), material and location (bottom left), plan (bottom right); dotted edges are realization.

Table 6: Conformance of the experiment to the LDL concept hierarchy.

concept	constraints	satisfied
experiment	4	4
comparative-experiment	17	17
controlled-experiment	3	3
hypothesis-discriminating-experiment	5	5
challenge-rescue-experiment	7	7

Table 7: Negative controls of the LDL concept layer.

modification	fails at	still an instance of
comparator dose 0.60 au	controlled-experiment	hypothesis-discriminating-experiment, comparative-experiment
control culture removed	comparative-experiment, controlled-experiment	hypothesis-discriminating-experiment
rescue given with the challenge (0 h)	challenge-rescue-experiment	hypothesis-discriminating-experiment, controlled-experiment
belief and answer removed	hypothesis-discriminating-experiment	controlled-experiment
rescue intervention missing from the plan	comparative-experiment	hypothesis-discriminating-experiment
comparator on another plate	comparative-experiment	hypothesis-discriminating-experiment
an observation step in two phases	comparative-experiment	hypothesis-discriminating-experiment
a well as part of a treatment	experiment, comparative-experiment	—

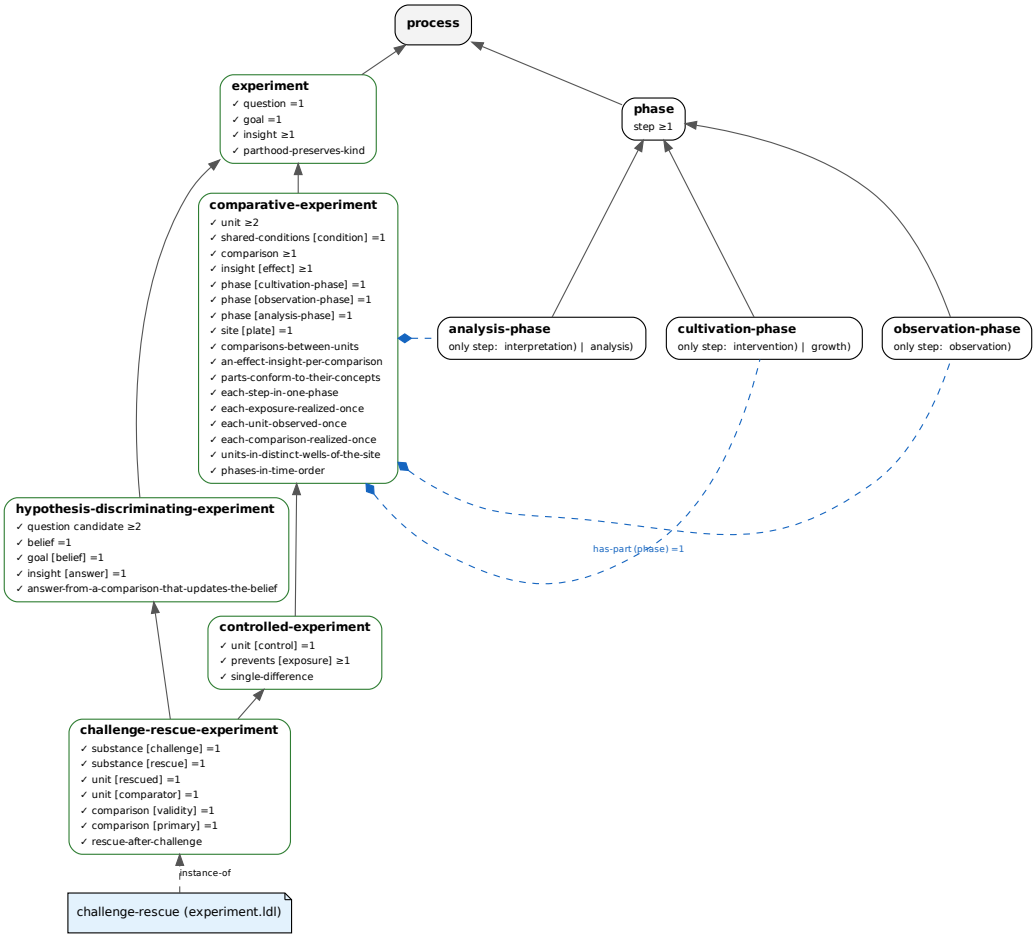


Figure 7: The LDL concept hierarchy with each concept's own constraints, checked against the experiment; the instance is declared `instance-of` challenge-rescue-experiment.

Table 8: Expressivity of the named checks of the LDL concept layer.

constraint	logic it needs
parthood-preserves-kind	OWL 2 DL
answer-from-a-comparison-that-updates-the-belief	OWL 2 DL
comparisons-between-units	SROIQ role inclusion
an-effect-insight-per-comparison	FOL
parts-conform-to-their-concepts	DL semantics
each-step-in-one-phase	OWL 2 DL
each-exposure-realized-once	OWL 2 DL + rule
each-unit-observed-once	FOL
each-comparison-realized-once	OWL 2 DL + rule
units-in-distinct-wells-of-the-site	FOL (path agreement)
phases-in-time-order	ALC(D) feature chains
single-difference	FOL with counting
rescue-after-challenge	ALC(D) feature chains

Table 9: The TBox by group.

group	axioms	definitions
Upper level	21	0
Relations and partonomy	41	0
Biological material and labware	24	5
Biological processes	7	0
Plans: designs, arms, administrations, comparisons	20	15
Factors, levels and treatments	14	4
Controls	6	4
Readouts and analysis plan	17	4
Design classes	10	10
Experiments, phases and steps	26	10
Data, hypotheses and conclusions	12	0
Laboratories, devices and capabilities	35	12
Design requirements	13	7

10 A first description-logic statement

The first DL statement (`experiment_dl.py`) was generated from the LDL concept file: each `has` clause became a qualified number restriction, each `only` clause a value restriction, a pattern's slots nested existentials, and `(role x)` the restriction $\exists \mathit{hasRole}.\mathit{XRole}$. Because it was generated, it could not drift from what the checker checks. Its main value was diagnostic: stating each named check in the logic it needs (Table 8) showed which parts of the experiment concept OWL 2 can carry and which it cannot. Single difference compares two individuals' sets of exposures and needs first-order logic with counting; "rescue after challenge" and the order of phases compare data values reached by different paths and need concrete domains with feature chains [23]; "all units on one plate" is a role-value map, undecidable together with the rest. The generator also verified that number restrictions are placed only on simple roles.

The first statement had three weaknesses that the second axiomatization removes: the design and the experiment were one entity, so a plan and its execution could not be told apart; roles were values of a *hasRole* slot rather than realizable entities borne by participants; and the concepts were *primitive*, so nothing was classified—the hierarchy was only checked against what was declared.

11 The OWL 2 DL axiomatization

The second axiomatization (`experiment_ontology.py`) states the framework as a biomedical ontology in $\mathcal{SROIQ}(\mathbf{D})$. It is written once, in a small DL syntax embedded in Python, and rendered three ways: as DL notation (this report and Appendix A), as OWL functional syntax (`experiment.ofn`) and as an Owlready2 ontology [25] over which HermiT [24] reasons. It has 246 TBox axioms over 174 concepts, 52 object properties and 12 data properties, in 13 groups (Table 9); 71 axioms are definitions (≡), which is what lets the reasoner classify. The namespace is `https://aicor.example/okf/experiment#`. This section presents the axioms that carry the framework; the full TBox is in Appendix A.

11.1 Upper level and partonomy

Continuants and occurrents are disjoint; material entities, specifically dependent continuants (qualities and realizable entities) and information content entities are disjoint; roles and dispositions are disjoint realizable entities; planned and biological processes are disjoint. Parthood is transitive and kind-preserving:
Trans(*hasPart*)

Process $\sqsubseteq \forall$*hasPart*.Process

MaterialEntity $\sqsubseteq \forall$*hasPart*.MaterialEntity

InformationContentEntity $\sqsubseteq \forall$*hasPart*.InformationContentEntity

containedIn $\circ$ *partOf* $\sqsubseteq$ *locatedIn*

hasPart $\circ$ *hasParticipant* $\sqsubseteq$ *hasParticipant*

The bearer relation ranges over specifically dependent continuants—qualities as well as roles and dispositions. An earlier version restricted it to realizable entities, which made organoid size, every size datum and the executed experiment's ABox unsatisfiable; HermiT found the error at the first run, and the negative controls were guarded so that they count only against a consistent base.

11.2 Biological material, agents and units

OrganoidCulture $\sqsubseteq$ MaterialEntity $\sqcap \exists$*hasOrganoid*.Organoid $\sqcap \exists$*hasMedium*.CultureMedium

Organoid $\sqsubseteq$ MaterialEntity $\sqcap \exists$*hasCell*.Cell

Well $\sqsubseteq$ MaterialEntity $\sqcap (\leq 1$*contains*.OrganoidCulture)

RecoveryCapacity $\sqsubseteq$ Disposition $\sqcap \exists$*inheresIn*.OrganoidCulture $\sqcap \forall$*realizedIn*.RecoveryProcess

ToxicityDisposition $\sqsubseteq$ Disposition $\sqcap \exists$*inheresIn*.ChemicalAgent $\sqcap \forall$*realizedIn*.InjuryProcess

Toxicant $\equiv$ ChemicalAgent $\sqcap \exists$*bearerOf*.ToxicityDisposition

ChallengeAgent $\equiv$ ChemicalAgent $\sqcap \exists$*bearerOf*.ChallengeAgentRole

ChallengeAgent $\sqsubseteq$ Toxicant

ExperimentalUnit $\equiv$ OrganoidCulture $\sqcap \exists$*bearerOf*.ExperimentalUnitRole

An organoid culture is organoids in medium; a well holds at most one culture. Agents are chemical entities defined by the roles they bear in the experiment; the challenge agent is a toxicant because its role is given for its toxicity disposition. An experimental unit is an organoid culture bearing the experimental-unit role. Recovery capacity and toxicity are dispositions realized only in recovery and injury processes, respectively.

11.3 Arms, defined by their treatment

An arm's class follows from what it administers. With the closure axioms of Section 12, the reasoner infers these classes from an arm's administration specifications alone:
UntreatedArm $\equiv$ ArmSpecification $\sqcap \neg \exists$*hasAdministrationSpec*.AdministrationSpecification

VehicleControlArm $\equiv$ ArmSpecification $\sqcap \exists$*hasAdministrationSpec*.VehicleSpec $\sqcap$ $\neg \exists$*hasAdministrationSpec*.(ChallengeSpec $\sqcup$ RescueSpec)

NegativeControlArm $\equiv$ UntreatedArm $\sqcup$ VehicleControlArm

ChallengeOnlyArm $\equiv$ ArmSpecification $\sqcap \exists$*hasAdministrationSpec*.ChallengeSpec $\sqcap$ $\neg \exists$*hasAdministrationSpec*.RescueSpec

RescueOnlyArm $\equiv$ ArmSpecification $\sqcap \exists$*hasAdministrationSpec*.RescueSpec $\sqcap$ $\neg \exists$*hasAdministrationSpec*.ChallengeSpec

ChallengeRescueArm $\equiv$ ArmSpecification $\sqcap \exists$*hasAdministrationSpec*.ChallengeSpec $\sqcap$ $\exists$*hasAdministrationSpec*.RescueSpec

ReplicatedArm $\equiv$ ArmSpecification $\sqcap (\geq 3$*hasAssignedUnit*.ExperimentalUnit)

PositiveControlArm $\equiv$ ArmSpecification $\sqcap \exists$*hasExpectedOutcome*.EvidenceBasedExpectation

11.4 Factors, levels and comparisons

A factor has at least two levels; an arm is at one level of each factor; a comparison contrasts the factors in which its arms differ. A comparison is classified by the arms it compares:
InjuryValidityComparison $\equiv$ ComparisonSpecification $\sqcap \exists$*compares*.ChallengeOnlyArm $\sqcap$ $\exists$*compares*.NegativeControlArm

RescueEffectComparison ≡ ComparisonSpecification ⊓ ∃*compares*.ChallengeRescueArm ⊓ ∃*compares*.ChallengeOnlyArm

AgentAloneComparison ≡ ComparisonSpecification ⊓ ∃*compares*.RescueOnlyArm ⊓ ∃*compares*.NegativeControlArm

RescueTimingComparison ≡ ComparisonSpecification ⊓ (= 2*compares*.ChallengeRescueArm)

Factor ⊑ PlanSpecification ⊓ (≥ 2*hasLevel*.FactorLevel)

FactorLevel ⊑ PlanSpecification ⊓ (= 1*levelOf*.Factor)

SingleFactorComparison ≡ ComparisonSpecification ⊓ (= 1*contrasts*.Factor)

InteractionContrast ⊑ PlanSpecification ⊓ (= 2*differenceOf*.ComparisonSpecification)

PositiveControlComparison ≡ ComparisonSpecification ⊓ ∃*compares*.PositiveControlArm ⊓ ∃*compares*.ChallengeOnlyArm

11.5 Readouts and the analysis plan

A readout measures a quality at a time and may be a proxy for another. The analysis plan has decision rules; a comparison rule applies to exactly one comparison; a deviation rule applies to none. A design is pre-specified when it has a readout, an analysis plan, and every comparison is covered by a rule:

ReadoutSpecification ⊑ PlanSpecification ⊓ ∃*measures*.Quality ⊓ ∃$atTime_h$.decimal

SizeReadout ≡ ReadoutSpecification ⊓ ∃*measures*.OrganoidSize ⊓ ∃*isProxyFor*.Viability

ComparisonRule ⊑ DecisionRule ⊓ (= 1*appliesTo*.ComparisonSpecification)

BayesianUpdateRule ⊑ ComparisonRule ⊓ (≥ 2*isAbout*.Hypothesis)

DeviationRule ⊑ DecisionRule ⊓ ¬∃*appliesTo*.⊤

CoveredComparison ≡ ComparisonSpecification ⊓ ∃$appliesTo^-$.ComparisonRule

PreSpecifiedDesign ≡ StudyDesign ⊓ ∃*hasReadout*.ReadoutSpecification ⊓ ∃*hasAnalysisPlan*.AnalysisPlan ⊓ ∀*hasComparisonSpec*.CoveredComparison

11.6 Design and experiment classes

Design classes are defined by arms, comparisons, factors and the analysis plan. Adequate control—the reviewer's standard—is the conjunction of agent control, a positive control, replication, single-factor comparisons and a pre-specified analysis. Experiment classes mirror design classes: an investigation executes exactly one design and is classified by it.

WellContrastedDesign ≡ StudyDesign ⊓ ∀*hasComparisonSpec*.SingleFactorComparison

ControlledDesign ≡ ComparativeDesign ⊓ ∃*hasArm*.NegativeControlArm

InjuryRescueDesign ≡ ControlledDesign ⊓ ∃*hasArm*.ChallengeRescueArm ⊓ ∃*hasArm*.ChallengeOnlyArm ⊓ ∃*hasComparisonSpec*.RescueEffectComparison

ValidatedInjuryRescueDesign ≡ InjuryRescueDesign ⊓ ∃*hasComparisonSpec*.InjuryValidityComparison

AgentControlledInjuryRescueDesign ≡ ValidatedInjuryRescueDesign ⊓ ∃*hasArm*.RescueOnlyArm ⊓ ∃*hasArm*.VehicleControlArm ⊓ ∃*hasComparisonSpec*.AgentAloneComparison

TherapeuticWindowDesign ≡ InjuryRescueDesign ⊓ (≥ 2*hasArm*.ChallengeRescueArm)

ReplicatedDesign ≡ StudyDesign ⊓ ∀*hasArm*.ReplicatedArm

AdequatelyControlledInjuryRescueDesign ≡ AgentControlledInjuryRescueDesign ⊓ PositiveControlledDesign ⊓ ReplicatedDesign ⊓ WellContrastedDesign ⊓ PreSpecifiedDesign

SufficientlyInformativeDesign ≡ StudyDesign ⊓ ∃$expectedInformationGain_b its$.$\text{decimal}_{\geq 0.5}$

Investigation ⊑ PlannedProcess ⊓ (= 1*executes*.StudyDesign)

ValidatedInjuryRescueExperiment ≡ Investigation ⊓ ∃*executes*.ValidatedInjuryRescueDesign

11.7 Phases, steps, data and hypotheses

A comparative experiment has exactly one cultivation, observation and analysis phase; each phase restricts its steps (administrations and incubations; assays; data transformations). A size datum is about organoid size and a proxy for viability. The recovery hypotheses are distinguished by what they posit:

ComparativeExperiment ≡ Investigation ⊓ ∃*executes*.ComparativeDesign

ComparativeExperiment $\sqsubseteq$ (= 1 *hasPhase*.CultivationPhase) $\sqcap$ (= 1 *hasPhase*.ObservationPhase) $\sqcap$ (= 1 *hasPhase*.AnalysisPhase)

CultivationPhase $\sqsubseteq$ Phase $\sqcap$ $\forall$ *hasStep*.(Administration $\sqcup$ Incubation)

SizeDatum $\sqsubseteq$ MeasurementDatum $\sqcap$ $\exists$ *isAbout*.OrganoidSize $\sqcap$ $\exists$ *isProxyFor*.Viability

GradedRecoveryHypothesis $\sqsubseteq$ RecoveryHypothesis $\sqcap$ $\neg\exists$ *posits*.LossOfRecoveryCapacity

ThresholdRecoveryHypothesis $\sqsubseteq$ RecoveryHypothesis $\sqcap$ $\exists$ *posits*.LossOfRecoveryCapacity

ProtectiveResponseHypothesis $\sqsubseteq$ ThresholdRecoveryHypothesis $\sqcap$ $\exists$ *posits*.(ProtectiveResponse $\sqcap$ $\exists$ *delays*.LossOfRecoveryCapacity)

11.8 Alignment to Cell 01's biology ontology

Where a concept means the same as one of Cell 01's biology ontology (`aicor-biology.ttl`, read only), it is subsumed by it: 17 axioms such as Organoid $\sqsubseteq$ bio : Organoid, ChallengeAdministration $\sqsubseteq$ bio : ControlledToxicChallenge and ImagingAssay $\sqsubseteq$ bio : ImagingProcess. Concepts whose meaning differs (the biology ontology's models are scientific models, ours are hypotheses that posit processes) are not aligned. Reasoning over both ontologies together is consistent with no unsatisfiable concept (HermiT 0.63 s).

12 Closure and the closed world

12.1 Closure axioms

OWL reasons under the open world. That an arm has a challenge and *no* rescue cannot be concluded from the absence of a rescue assertion, and the definitions of Section 11 contain negations and universal restrictions. The ABox therefore closes what the design states: every arm has at most as many administrations and assigned units as it lists, the design at most as many arms and comparisons, and each comparison at most as many contrasted factors. Together with distinctness assertions, these let HermiT conclude, for example, that the comparator arm—one administration, which is a challenge specification, disjoint from rescue specifications—is a challenge-only arm, and that a design all of whose (closed) arms are replicated satisfies $\forall$ *hasArm*.ReplicatedArm. The closure of the design's arms had at first been stated over arm specifications only; under the open world another kind of filler remained possible and the universal restriction could not be concluded, so the closure now ranges over every filler.

12.2 Closed-world checks

What OWL 2 cannot state, or cannot conclude under the open world, is checked by queries over the design as given: every arm has one level of each factor; each comparison contrasts exactly one factor; the administration schedules of compared arms differ only at time points belonging to the contrasted factor (vehicle matching); whether the design is a full factorial (information, not a requirement); whether some interaction is estimable (a complete 2×2 sub-table of two factors' levels); every comparison has a decision rule; a positive control licenses the primary comparison; at least 3 replicates per arm (an ILLUSTRATIVE criterion); units off the plate edge; all units on one plate; and, for an executed experiment, each administration specification executed exactly once. The single-factor check replaces the earlier time-point check: the therapeutic-window comparison (late vs. early rescue) differs at two time points but in one factor, the time of rescue.

13 Reasoning results

13.1 Consistency

HermiT finds the ontology consistent with the current experiment's ABox (0.57 s), with the improved design's (0.74 s), with both together, and with Cell 01's biology ontology, with no unsatisfiable concept in any case.

13.2 The current experiment

Table 10 lists the most specific concepts the reasoner infers for the design, its arms and comparisons, the experiment and some steps. Nothing in the table is asserted: the ABox states only administrations, assignments, comparisons, levels, rules and closures.

The design is a *validated injury–rescue design*: it has a negative-control arm, a challenge–rescue and a challenge-only arm, a rescue-effect and an injury-validity comparison. It is also factorial (two factors, challenge and rescue), well

Table 10: Realization of the current experiment (HermiT).

individual	inferred most specific concepts
challenge_rescue_design	FactorialDesign, HypothesisDiscriminatingDesign, PreSpecifiedDesign, RequiresImaging, SufficientlyInformativeDesign, ValidatedInjuryRescueDesign, WellContrastedDesign
objective	DiscriminationObjective
agent_challenge	ChallengeAgent
agent_rescue	RescueAgent
arm_rescued	ChallengeRescueArm
arm_comparator	ChallengeOnlyArm
arm_control	UntreatedArm
cmp_validity	CoveredComparison, InjuryValidityComparison, SingleFactorComparison
cmp_primary	CoveredComparison, RescueEffectComparison, SingleFactorComparison
experiment_challenge_rescue	FactorialExperiment, HypothesisDiscriminatingExperiment, ValidatedInjuryRescueExperiment
step_x1	ChallengeAdministration
step_x3	RescueAdministration
step_a1	ModelUpdate, StatisticalComparison

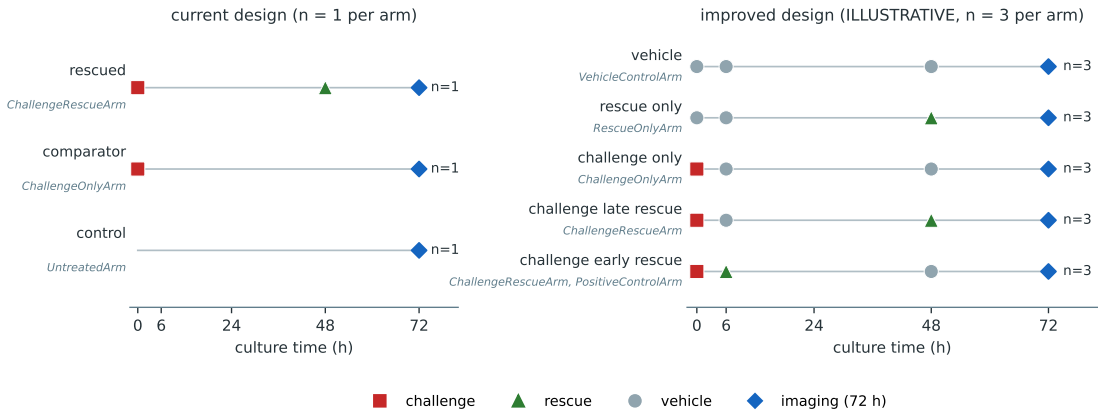


Figure 8: The two designs on a time axis: administrations of challenge, rescue and vehicle, and imaging at 72 h; under each arm, the classes HermiT infers for it.

contrasted (each comparison contrasts one factor), pre-specified (both comparisons have a rule: a threshold rule for validity, a Bayesian update for the primary comparison), hypothesis-discriminating and sufficiently informative. It is *not* agent-controlled (no rescue-only arm, no vehicle arm), not positive-controlled and not replicated, and therefore not adequately controlled. The experiment that executes it is classified accordingly.

13.3 An improved design

The improved design (ILLUSTRATIVE, not run) answers the reviewer: it is a factorial of *challenge* (vehicle, 0.85 au at 0 h) and *time of rescue* (none, 6 h, 48 h), every level vehicle-matched, with 5 of the six combinations as arms (Table 11). The early-rescue arm is a positive control: its outcome is expected from prior evidence (rescue at 6 h was effective at every dose tested), so that a null result at 48 h means that the late rescue fails, not that the agent or the assay failed in this run. Each arm has three units in randomised interior wells of one plate, and the analysis plan has a rule for each of the 5 comparisons and a deviation rule. Its expected information gain for the primary contrast is 0.641 bit; the surrogate does not model what the control arms add.

The reasoner classifies the improved design as an *adequately controlled injury–rescue design*, a therapeutic-window design (two challenge–rescue arms), interaction-aware and deviation-aware (Table 13). The early-rescue arm is

Table 11: Arms of the improved design (ILLUSTRATIVE).

arm	challenge	rescue time	schedule	n
vehicle	vehicle	none	vehicle 0 au at 0 h; vehicle 0 au at 6 h; vehicle 0 au at 48 h	3
rescue only	vehicle	h48	vehicle 0 au at 0 h; vehicle 0 au at 6 h; rescue 0.7 au at 48 h	3
challenge only	c085	none	challenge 0.85 au at 0 h; vehicle 0 au at 6 h; vehicle 0 au at 48 h	3
challenge late rescue	c085	h48	challenge 0.85 au at 0 h; vehicle 0 au at 6 h; rescue 0.7 au at 48 h	3
challenge early rescue	c085	h6	challenge 0.85 au at 0 h; rescue 0.7 au at 6 h; vehicle 0 au at 48 h	3

Table 12: Comparisons of the improved design.

comparison	arms	contrasts	rule	licensed by
validity	challenge only vs vehicle	challenge	threshold	—
primary	challenge late rescue vs challenge only	rescue_time	bayes	validity→primary, positive_control→primary
agent_alone	rescue only vs vehicle	rescue_time	estimate	—
positive_control	challenge early rescue vs challenge only	rescue_time	threshold	—
window	challenge late rescue vs challenge early rescue	rescue_time	estimate	validity→window

Table 13: Realization of the improved design (HermiT).

individual	inferred most specific concepts
`improved_design`	AdequatelyControlledInjuryRescueDesign, DeviationAwareDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, RequiresImaging, SufficientlyInformativeDesign, TherapeuticWindowDesign
`improved_objective`	DiscriminationObjective
`agent_challenge`	ChallengeAgent
`agent_rescue`	RescueAgent
`agent_vehicle`	Vehicle
`improved_arm_vehicle`	ReplicatedArm, VehicleControlArm
`improved_arm_rescue_only`	ReplicatedArm, RescueOnlyArm
`improved_arm_challenge_only`	ChallengeOnlyArm, ReplicatedArm
`improved_arm_challenge_late_rescue`	ChallengeRescueArm, ReplicatedArm
`improved_arm_challenge_early_rescue`	ChallengeRescueArm, PositiveControlArm, ReplicatedArm
`improved_cmp_validity`	CoveredComparison, InjuryValidityComparison, SingleFactorComparison
`improved_cmp_primary`	CoveredComparison, RescueEffectComparison, SingleFactorComparison
`improved_cmp_agent_alone`	AgentAloneComparison, CoveredComparison, SingleFactorComparison
`improved_cmp_positive_control`	CoveredComparison, PositiveControlComparison, RescueEffectComparison, SingleFactorComparison
`improved_cmp_window`	CoveredComparison, RescueTimingComparison, SingleFactorComparison

both a challenge–rescue arm and a positive-control arm; the positive-control comparison is also a rescue-effect comparison, which is correct; and the late-versus-early comparison is a rescue-timing comparison—the contrast that asks directly whether the rescue's window closes.

13.4 Closed-world checks

Table 14 lists the closed-world checks for both designs. The current design passes the single-factor, schedule-matching, coverage-by-rules and placement-on-one-plate checks and the execution check, and fails on interaction estimability (the rescue-only cell is missing), the positive control, replication and edge placement. The improved design passes every required check; that it is not a full factorial (the vehicle–early-rescue cell is absent) is reported as information, because no comparison needs that cell.

13.5 Variants

To check that adequacy is not classified by accident, three variants of the improved design each remove one ingredient. Each loses exactly the corresponding class and fails exactly the corresponding closed-world check (Table 15).

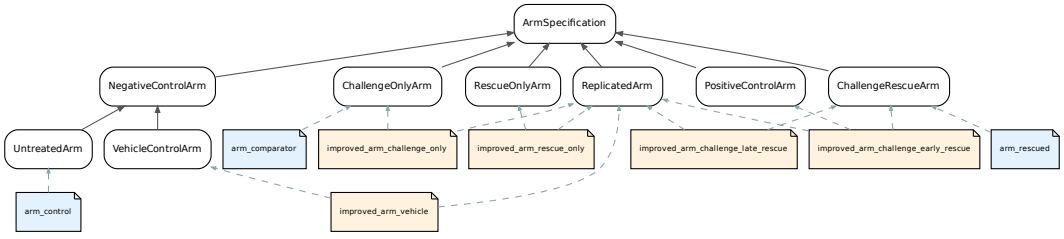


Figure 9: Inferred taxonomy of arms, with the arms of the current (blue) and improved (orange) designs realized under it.

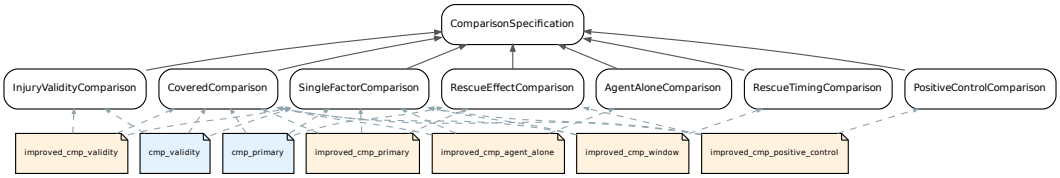


Figure 10: Inferred taxonomy of comparisons, with both designs' comparisons.

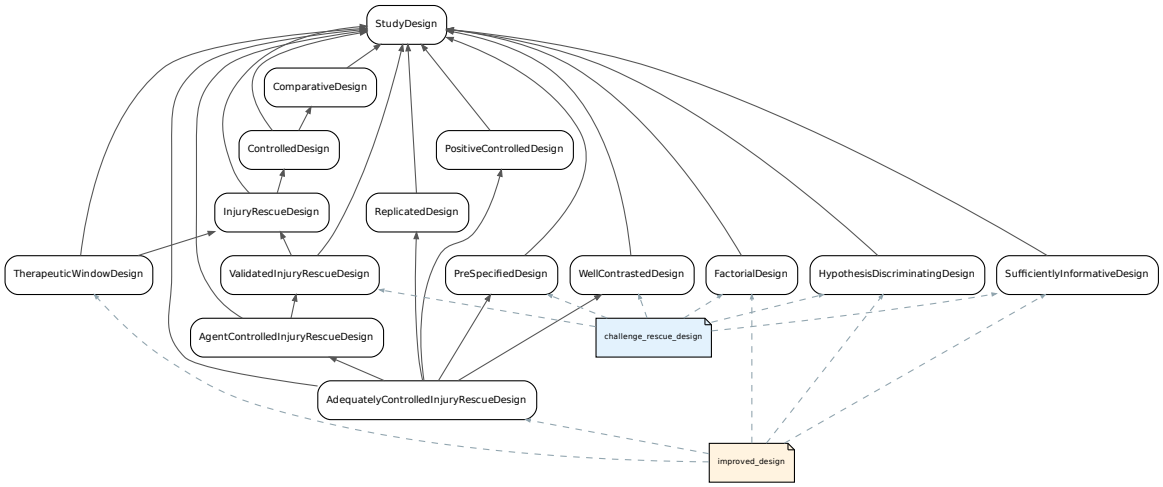


Figure 11: Inferred taxonomy of designs, with both designs. Solid edges are subsumptions computed by HermiT; dashed edges are realizations.

Table 14: Closed-world checks.

check	current	improved	detail (current, else improved)
every arm has one level of every factor	pass	pass	3 arms × 2 factors
single factor: validity	pass	pass	comparator vs control differ in challenge
schedules matched: validity	pass	pass	administrations differ at 0 h, all belonging to the contrasted factor
single factor: primary	pass	pass	rescued vs comparator differ in rescue
schedules matched: primary	pass	pass	administrations differ at 48 h, all belonging to the contrasted factor
full factorial (information, not required)	*info*	*info*	3 of 4 level combinations; missing: challenge=none, rescue=0p7au_48h
interaction estimable	fail	pass	no complete 2×2 sub-table
every comparison has a decision rule	pass	pass	2 of 2
positive control licenses the primary comparison	fail	pass	no positive-control arm
replicates ≥ 3 per arm	fail	pass	rescued 1, comparator 1, control 1
units off the plate edge	fail	pass	edge wells: ['A01', 'A02', 'A05']
all units on one plate	pass	pass	plate_P17
each administration specification executed once	pass	—	3 administrations for 3 specifications
single factor: agent_alone	—	pass	rescue_only vs vehicle differ in rescue_time
schedules matched: agent_alone	—	pass	administrations differ at 48 h, all belonging to the contrasted factor
single factor: positive_control	—	pass	challenge_early_rescue vs challenge_only differ in rescue_time
schedules matched: positive_control	—	pass	administrations differ at 6 h, all belonging to the contrasted factor
single factor: window	—	pass	challenge_late_rescue vs challenge_early_rescue differ in rescue_time
schedules matched: window	—	pass	administrations differ at 6 h, 48 h, all belonging to the contrasted factor

Table 15: Variants of the improved design.

variant	most specific design concepts	closed-world failures
improved without the positive-control arm	AgentControlledInjuryRescueDesign, DeviationAwareDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, ReplicatedDesign, RequiresImaging, SufficientlyInformativeDesign, WellContrastedDesign	positive control licenses the primary comparison
improved with one unit per arm	AgentControlledInjuryRescueDesign, DeviationAwareDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, PositiveControlledDesign, RequiresImaging, SufficientlyInformativeDesign, TherapeuticWindowDesign, WellContrastedDesign	replicates ≥ 3 per arm
improved without a pre-specified analysis	AgentControlledInjuryRescueDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, PositiveControlledDesign, ReplicatedDesign, RequiresImaging, StudyDesign, SufficientlyInformativeDesign, TherapeuticWindowDesign, WellContrastedDesign	every comparison has a decision rule

Table 16: Negative controls of the OWL axiomatization.

modification	HermiT
a well as part of an arm (parthood crosses kinds)	inconsistent, as required
a culture assigned to two arms (assignedTo is functional)	inconsistent, as required
an administration added to the untreated arm (closure)	inconsistent, as required
the graded hypothesis posits a loss of recovery capacity	inconsistent, as required
the challenge agent also bears the rescue role (disjoint agents)	inconsistent, as required
one decision rule applied to two comparisons	inconsistent, as required
an incubation function realized in an imaging assay	inconsistent, as required
a factor level that is also an arm (disjoint plan parts)	inconsistent, as required

13.6 Negative controls and inferred subsumptions

8 modifications must each make the ontology inconsistent; all 8 do (Table 16). They exercise kind-preserving parthood, the functionality of assignment, arm closure, the disjointness of what hypotheses posit, the disjointness of agents, the cardinality of comparison rules, the realization of device functions (Section 14) and the disjointness of plan parts.

The reasoner also derives subsumptions that no axiom states; the 30 found are AdequatelyControlledInjuryRescueDesign $\sqsubseteq$ RequiresDosing; AdequatelyControlledInjuryRescueExperiment $\sqsubseteq$ ValidatedInjuryRescueExperiment; AgentControlledInjuryRescueDesign $\sqsubseteq$ RequiresDosing; ComparativeDesign $\sqsubseteq$ RequiresCultivation; ControlledDesign $\sqsubseteq$ RequiresCultivation; ControlledExperiment $\sqsubseteq$ ComparativeExperiment; DeviationAwareDesign $\sqsubseteq$ RequiresCultivation; FactorialDesign $\sqsubseteq$ RequiresCultivation; HypothesisDiscriminatingDesign $\sqsubseteq$ RequiresCultivation; InjuryRescueDesign $\sqsubseteq$ RequiresDosing; InjuryRescueExperiment $\sqsubseteq$ ControlledExperiment; InteractionAwareDesign $\sqsubseteq$ RequiresCultivation; NegativeControlArm $\sqsubseteq$ ArmSpecification; PositiveControlledDesign $\sqsubseteq$ RequiresCultivation; PreSpecifiedDesign $\sqsubseteq$ RequiresCultivation; ReplicatedDesign $\sqsubseteq$ RequiresCultivation; RequiresAbsorbance $\sqsubseteq$ RequiresEndpointHandling; RequiresDosing $\sqsubseteq$ RequiresCultivation; RequiresEndpointHandling $\sqsubseteq$ RequiresCultivation; RequiresImaging $\sqsubseteq$ RequiresCultivation; RequiresLuminescence $\sqsubseteq$ RequiresEndpointHandling; RequiresMultipleEnvironments $\sqsubseteq$ RequiresCultivation; SufficientlyInformativeDesign $\sqsubseteq$ RequiresCultivation; TherapeuticWindowDesign $\sqsubseteq$ RequiresDosing; TherapeuticWindowExperiment $\sqsubseteq$ InjuryRescueExperiment; UntreatedArm $\sqsubseteq$ NegativeControlArm; ValidatedInjuryRescueDesign $\sqsubseteq$ RequiresDosing; ValidatedInjuryRescueExperiment $\sqsubseteq$ InjuryRescueExperiment; VehicleControlArm $\sqsubseteq$ NegativeControlArm; WellContrastedDesign $\sqsubseteq$ RequiresCultivation. In particular the whole chain of experiment classes—adequately controlled, validated, injury–rescue, controlled, comparative—is inferred from the definitions of the design classes they mirror, and the two negative-control arm classes are recognised as specializations of their disjunction. The remaining subsumptions place design classes under the requirement classes of Section 14: whatever has an arm requires cultivation, and every injury–rescue design requires dosing.

14 What our laboratory affords

Biomedical laboratories are not general-purpose. What experiments a laboratory affords is fixed by its equipment, the environments it controls, its containment, the assays it has validated and the reagents it keeps; a cell-culture laboratory, an imaging core, a plate-reader core and a hypoxia facility afford different experiments, and most experiments of interest need more than one of them. Cell 01 is a *treatment-and-imaging module*: it incubates, doses, images, calibrates and quarantines plates, and a clean-sample robot connects it to the rest of the site. A robot scientist that proposes designs therefore has to know, for each design, whether it can be run here, which of its steps have to be performed elsewhere, and what is missing. This section adds capability profiles of laboratories to the ontology, classifies designs by what they require, and matches the two (`rsa/lab_affordance.py`).

14.1 Capabilities as device functions

Following BFO, a capability is modelled as a *function*: a disposition that inheres in a device and is realized only in processes of one kind. A liquid handler bears a dosing function, realized only in administrations; an imager bears an imaging function, realized only in imaging assays. A laboratory has its devices as parts, and its capability classes are *defined* by the functions its devices bear, by the number of controlled environments it provides and by its validated readout bindings (a binding is information about which quality a device's readout measures, and at which validation level). The reasoner therefore classifies a laboratory from its devices, and nothing about what a laboratory can do is asserted directly. Selected axioms:

Laboratory $\sqsubseteq$ MaterialEntity $\sqcap$ $\exists$*hasDevice*.Device

DeviceFunction $\sqsubseteq$ Disposition $\sqcap$ $\exists$*inheresIn*.Device

DosingFunction $\sqsubseteq$ DeviceFunction $\sqcap$ $\forall$*realizedIn*.Administration

BrightfieldImagingFunction $\sqsubseteq$ DeviceFunction $\sqcap$ $\forall$*realizedIn*.ImagingAssay

DosingLab $\equiv$ Laboratory $\sqcap$ $\exists$*hasDevice*.$\exists$*bearerOf*.DosingFunction

MultiEnvironmentLab $\equiv$ Laboratory $\sqcap$ ($\geq$ 2*providesEnvironment*.EnvironmentProfile)

ValidatedSizeReadoutLab $\equiv$ Laboratory $\sqcap$ $\exists$*hasReadoutBinding*.(ValidatedReadoutBinding $\sqcap$ $\exists$*isAbout*.OrganoidSize)

TreatmentAndImagingModule $\equiv$ DosingLab $\sqcap$ CultivationLab $\sqcap$ BrightfieldImagingLab $\sqcap$ ValidatedSizeReadoutLab

Reagent stocks are located in a laboratory (*stocks*), not parts of it, and an environment profile is a plan specification that a laboratory realizes, so that two incubators running at the same setpoints count as one environment. Because a function is realized only in processes of its kind, a function realized in the wrong kind of process is a contradiction: an incubation function realized in an imaging assay is the eighth negative control of Table 16.

14.2 Design requirements

What a design needs from a laboratory follows from its content, so requirement classes are defined as well: a design with an administration requires dosing, one with an imaging readout requires imaging, one with an endpoint readout (luminescence, absorbance) requires the corresponding reader and endpoint handling, and one with an environment factor requires as many controlled environments as the factor has levels.

RequiresDosing $\equiv$ StudyDesign $\sqcap$ $\exists$*hasArm*.$\exists$*hasAdministrationSpec*.AdministrationSpecification

RequiresImaging $\equiv$ StudyDesign $\sqcap$ $\exists$*hasReadout*.ImagingReadout

RequiresLuminescence $\equiv$ StudyDesign $\sqcap$ $\exists$*hasReadout*.LuminescenceReadout

RequiresEndpointHandling $\equiv$ StudyDesign $\sqcap$ $\exists$*hasReadout*.EndpointReadout

RequiresMultipleEnvironments $\equiv$ StudyDesign $\sqcap$ $\exists$*hasFactor*.EnvironmentFactor

Each requirement is met by one capability class (Table 17). This correspondence is a table, not an axiom: it relates classes of different categories (designs and laboratories), and stating it as a subsumption would make every design a laboratory. The reasoner adds requirements no axiom states for a design class: every injury–rescue design, validated or adequately controlled, requires dosing; every design that has an arm requires cultivation; and luminescence and absorbance readouts require endpoint handling (Section 13).

14.3 Cell 01's profile

Cell 01's profile is not written by hand; it is read from three sources in the repository. The ten component records list each device's capability verbs, which a table maps to device functions (Table 18); the device registry lists the reagent stocks and the readout bindings with their validation level; and the twin's measured durations give the time a dosing

Table 17: Requirements and the capabilities that meet them.

design requirement	met by laboratory capability
RequiresDosing	DosingLab
RequiresCultivation	CultivationLab
RequiresImaging	BrightfieldImagingLab
RequiresLuminescence	LuminescenceReadoutLab
RequiresAbsorbance	AbsorbanceReadoutLab
RequiresMultipleEnvironments	MultiEnvironmentLab

Table 18: Cell 01's devices, their capability verbs and the functions they are evidence of.

device (component record)	capability verbs	functions
H4 Quarantine Station	isolate_sample, retain_invalid_run, prevent_reentry, release_by_authorization	QuarantineFunction
L09 Automated Plate Hotel	buffer_plate, store_plate, retrieve_plate, report_inventory, reserve_slot, verify_identity	StorageFunction
L10 Automated CO2 Incubator	incubate_plate, maintain_environment, store_plate, release_plate, report_environment, raise_excursion_alarm	IncubationFunction, StorageFunction
L12 Automated Liquid-Handling Station	pick_tip, detect_liquid_surface, aspirate, dispense, mix, dose_well, eject_tip	DosingFunction
L17 Bright-Field Imaging Station	load_plate, autofocus, image_well, image_plate_region, compute_focus_quality, export_raw_image	BrightfieldImagingFunction
L21 Quality-Control and Calibration Station	run_reference_measurement, validate_imager, validate_barcode, validate_tool_frame	CalibrationFunction
R01 Fixed Precision Plate-Handling Arm	pick_sbs_plate, place_sbs_plate, handover_plate, reseat_plate, inspect_grasp, move_plate_between_local_stations	PlateHandlingFunction
R02 Rail-Mounted Flexible Manipulation System	translate_workspace, pick_plate, pick_tube, handle_rack, press_button, turn_selector, open_access_door, change_tool, inspect_station	—
R03 Clean Sample AMR with H2 Deterministic Dock	navigate, transport_clean_sample, dock_precisely, handover_carrier, opportunity_charge	TransportFunction
R02 Automatic Tool Changer and Tool Magazine	dock_tool, lock_tool, verify_tool, unlock_tool, store_tool, select_tool_for_task	—

session takes. Verbs without a function (aspirate, autofocus, press a button) are steps of the functions' realizations, not capabilities of their own, and are not mapped.

Cell 01 provides one controlled environment (37 °C, 5 % CO_2, 90 % humidity, from the incubator's record), stocks the agents challenge, rescue, and has one validated readout binding: the imaging station's organoid radius, a readout of organoid size. A plate may be outside the incubator for at most 600 s per excursion; a dosing session costs about 205 s of transfers (two legs of each robot arm, retrieval, storage, tool coupling) and 35.6 s per well, so at most **11 wells** can be dosed per session (these durations are the twin's, ILLUSTRATIVE).

HermiT classifies the laboratories in 1.76 s, with both laboratories and all four designs in one ontology (Table 19). Cell 01 is a treatment-and-imaging module; it is not an endpoint-reader, metabolic-readout or multi-environment laboratory, because it has no endpoint reader, no validated metabolic readout and one environment. A second laboratory, a site core facility with a multimode plate reader and a dock for the clean-sample robot, is ILLUSTRATIVE (not modelled from records); it stands in for the facilities to which steps that Cell 01 cannot perform are allocated.

Table 19: Capability classes of the two laboratories (HermiT).

laboratory	is (HermiT)	is not
Cell 01 (automated treatment-and-imaging module)	BrightfieldImagingLab, ConnectedLab, ContainmentCapableLab, CultivationLab, DosingLab, QualityControlledLab, TreatmentAndImagingModule, ValidatedSizeReadoutLab	AbsorbanceReadoutLab, LuminescenceReadoutLab, MultiEnvironmentLab, ValidatedMetabolicReadoutLab
Core facility: multimode plate reader (ILLUSTRATIVE)	AbsorbanceReadoutLab, ConnectedLab, LuminescenceReadoutLab, ValidatedMetabolicReadoutLab	BrightfieldImagingLab, ContainmentCapableLab, CultivationLab, DosingLab, MultiEnvironmentLab, QualityControlledLab, TreatmentAndImagingModule, ValidatedSizeReadoutLab

Table 20: What Cell 01 and the site afford: four designs (ILLUSTRATIVE).

design	requires (HermiT)	verdict	missing	with reagents stocked
the current challenge-rescue design	Cultivation, Dosing, Imaging	executable in Cell 01	—	—
the improved design (vehicle, rescue-only, early rescue, 3 replicates)	Cultivation, Dosing, Imaging	not executable as modelled	reagent not stocked in Cell 01: vehicle	executable in Cell 01
the improved design with an ATP luminescence endpoint at 72 h	Cultivation, Dosing, EndpointHandling, Imaging, Luminescence	not executable as modelled	reagent not stocked in Cell 01: vehicle	executable on site with transfers
a challenge x oxygen design (environment factor)	Cultivation, Dosing, Imaging, MultipleEnvironments	not executable as modelled	RequiresMultipleEnvironments: no laboratory is a MultiEnvironmentLab	—

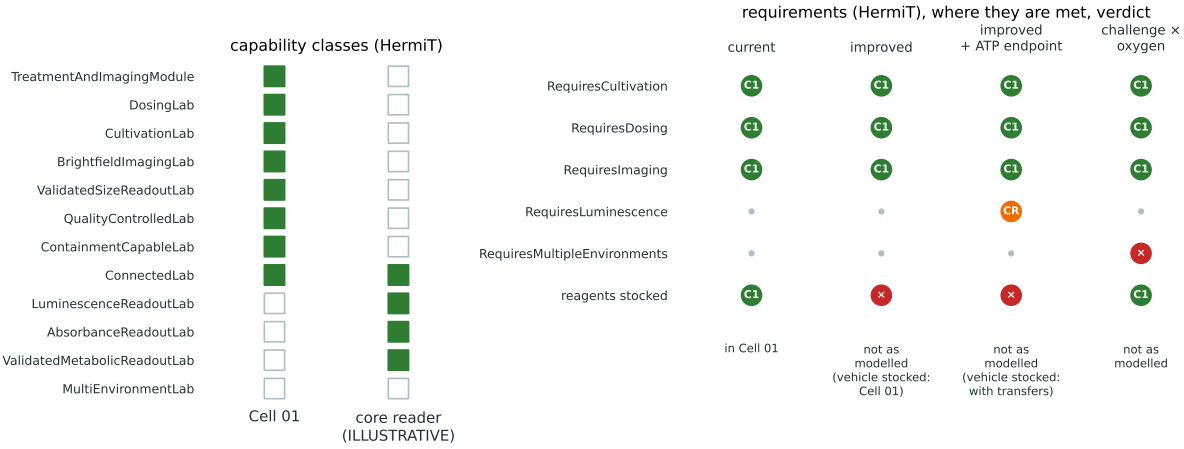


Figure 12: Left: capability classes HermiT infers for the two laboratories. Right: the requirements of the four designs and the laboratory each is allocated to (C1 Cell 01, CR core reader, × unmet), with the verdict below.

14.4 Matching designs against laboratories

Whether a design is executable is a closed-world question over the individuals: it concerns the laboratories and stocks the site has, not those it might have. For each design, the matcher (i) takes the requirement classes HermiT infers for it and allocates each to a laboratory with the meeting capability, Cell 01 first; (ii) checks that every agent the design administers is stocked where the dosing happens; (iii) checks that each readout has a validated binding to the quality it is about in the allocated laboratory; (iv) counts the environments an environment factor needs; (v) splits the doses due at each time point into sessions that fit the excursion window; (vi) checks the units against the 60 interior wells of one plate; (vii) checks that destructive endpoint readouts come after every imaging; and (viii) checks that a transfer to every other allocated laboratory is possible (both connected). The verdict is *executable in Cell 01*, *executable on site with transfers*, or *not executable as modelled* with what is missing; when only reagents are missing, the matcher also reports the verdict with them stocked.

14.5 Four designs

The current design requires cultivation, dosing and imaging, all met by Cell 01, with both agents stocked and a validated size readout; its three doses (two at 0 h, one at 48 h) take one session each. It is *executable in Cell 01*.

The improved design has the same requirements, but administers a vehicle that Cell 01 does not stock, so it is *not executable as modelled*; with the vehicle stocked it would be *executable in Cell 01*. Its 15 units receive 15 administrations at each of 0, 6 and 48 h, more than the 11 that fit one excursion, so each time point needs two sessions and the last wells are dosed about ten minutes after the first. Against a rescue delay of 6 or 48 h this is small, but it is systematic if the arms are dosed in a fixed order; the sessions should interleave the arms, which is the same randomisation that the placement check asks for on the plate.

Table 21: Existing standards and the framework.

standard	covers	layers	used here as	gap
ISA [17]	investigation, study, assay; factors and factor values per sample	design, data	units, factors, assays	no comparisons, controls by purpose, adequacy, reasoning
OBI [14]	planned processes, study designs, assays, roles; BFO	all five	the style of the upper level, roles, assays	this work adds defined design classes and adequacy
EXPO [11]	investigations, hypotheses, designs, results	epistemic, design	hypothesis–design–result link	no defined control or comparison types
CDISC SDM [10]	arms, epochs, cells, activities (clinical)	design	arms × epochs = our arms × phases	clinical only; no logic
EXACT [12], LabOP, Autoprotocol	protocol steps, parameters, containers	protocol, execution	our lab designators and CRAM plans	no design layer
ARRIVE [6], CONSORT [7]	reporting checklists	design (reported)	replication, randomisation, blinding, controls	checked by humans after the fact

The improved design with an ATP endpoint adds a luminescence readout of metabolic activity at 72 h, after the imaging. HermiT classifies it as requiring luminescence and endpoint handling, which no device of Cell 01 offers; the step is allocated to the core reader, which has a validated binding for it, and the transfer is possible because both laboratories are connected by the clean-sample robot. With the vehicle stocked it is *executable on site with transfers*. The time the plate spends outside the controlled environment on the way is not modelled; since the ATP read is the last step and destructive, it does not affect the imaging readout.

The challenge × oxygen design varies the environment (normoxia, hypoxia) as a factor. It requires 4 capabilities, among them MultiEnvironmentLab: two controlled environments, of which Cell 01 provides one and the core facility none. It is *not executable as modelled*; running it would need a second incubator at a different oxygen setpoint, or a hypoxia facility and transfers.

14.6 What the scope is, and what it is not

Within the modelled designs, Cell 01 affords in-vitro experiments on cultures in one plate, in one environment, with treatments by the agents it stocks and imaging of organoid size as the readout: injury, rescue and agent-control designs, therapeutic-window designs and their factorials, up to 11 administrations per session and 60 interior wells per plate. It affords designs with endpoint assays only together with a core facility, and designs with environment factors not at all. For the adequately controlled design of Section 13 the only missing resource is a vehicle stock.
Three limitations bound this result. The map from capability verbs to functions is hand-written, and the general manipulation verbs of the rail-mounted robot are not mapped, so capabilities that need a tool change are not represented. The core facility, the vehicle and the durations are ILLUSTRATIVE, and transfer times between laboratories are not modelled. And what can be dosed, imaged or read is classified, but whether an assay is suitable for a question (sensitivity, dynamic range) is not: that is a property of a validated binding, which is recorded as a level and not reasoned about.

15 Relation to existing standards

Table 21 relates the framework to existing standards. None of them combines the three things the robot scientist needs: a design layer with comparisons and controls defined by purpose, a reasoner that classifies designs from their content, and a link to the plans the robot executes. OBI is the closest in spirit and would be the natural upper ontology to import; the reporting guidelines list many of the criteria that the closed-world layer checks, but for human authors and after the fact.

16 Discussion

What the representation buys. The final representation turns a reviewer's judgement into a computation before the experiment is run. That the current design lacks agent and positive controls and replication is not a comment in a report but the absence of a class in a realization; that the improved design has them is a derived fact with a proof (the reasoner's), not an assertion. The same machinery explains a design: the classes an arm or a comparison belongs to say what it is for. And because the actions and their order are derived from the design, a change of design—a new arm, a new comparison—changes the plan the robot executes without hand editing.

Open and closed world. The division of labour between OWL and the closed-world queries is not a workaround but a consequence of the questions asked. "What kind of design is this?" is a question about what follows from what is stated, which the open world answers soundly once the design's own extent is closed. "Does this design contain three replicates per arm?" is a question about the design as given, which only the closed world answers. Making the closure explicit in the ABox, rather than switching to closed-world semantics wholesale, keeps the classification monotonic: adding an arm can only add classes.

Defined versus primitive concepts. The LDL concept layer checked an experiment against the concept it declared; the OWL layer *finds* the concepts an experiment belongs to. The difference matters for a robot scientist that generates designs: it should not need to know in advance what kind of experiment it has produced.

The proxy and the posit. Two modelling decisions carry the D016 discipline into the ontology. Data are proxies: the size datum is about organoid size and a proxy for viability; that viability is what the experiment is about is visible, and so is the assumption. Hypotheses posit: the threshold hypothesis posits a loss of recovery capacity; the graded one is defined by not positing it, and a graded hypothesis that posits it is inconsistent. The ontology can thus represent competing biology without committing to any of it.

The case study as a test. The case is small, but it exercises every layer: an epistemic objective with an information target, a design with comparisons of different purposes, a protocol refined into lab actions, an execution with an audit trail, and a hidden biological state reached through a proxy. The reviewer's critique of Section 4 served as an independent specification of what the representation should detect, and it does.

17 Limitations

- **Not executed.** The CRAM/RPL plans are generated and checked but not executed; no Lisp CRAM is installed. The surrogate run is a simulation, and its numbers are ILLUSTRATIVE (D016).
- **Two layers not yet joined.** Factors, levels, positive controls and the analysis plan exist in the OWL layer; the authored `experiment.ldl`, the LDL concept checker and the CRAM plans do not have them yet. The current design's factors are read off its schedules.
- **Upper ontology in style only.** BFO and OBI are followed in style, in an own namespace, not imported; the alignment to Cell 01's biology ontology is by subsumption where the meaning is the same.
- **Illustrative criteria.** The minimum of three replicates, the 0.5 bit information target and the validity criterion (an effect below zero by more than two standard deviations) are placeholders, not recommendations.
- **Coverage.** The taxonomy covers in-vitro injury–rescue experiments. Positive controls are defined; randomisation is checked only as placement off the plate edge; blinding, blocking, technical versus biological replicates, dose–response and time-course designs, and the in vivo and clinical branches are not axiomatized.
- **Information of control arms.** The information-gain model covers the primary contrast; it does not represent what control arms add (the confounds they exclude are outside the surrogate).
- **Laboratory profiles.** Capability verbs are mapped to device functions by a hand-written table; the rail-mounted robot's general manipulation is not mapped; the core facility, the vehicle and the durations are ILLUSTRATIVE; transfer times between laboratories, assay suitability and costs are not modelled.
- **Reproducibility pipeline.** The generators of this report are not yet stages of `rsa.reproduce`; adding them would change the manifest hash quoted by the companion report and paper.
- **References.** The references were compiled without database access and must be checked before publication.

18 Roadmap

1. **One source.** Extend `experiment.ldl` with factors, levels, readouts and the analysis plan; generate the OWL ABox, the culture plan and the CRAM plans from it, so that every layer derives from one authored design.
2. **Adequacy in the loop.** Make the reasoner's classification a constraint of the design planner: among the designs that reach the information target, choose the cheapest *adequately controlled* one, and report the classes a cheaper design would lose.

3. **Affordance in the loop.** Restrict the design planner to designs the site affords, and report for each rejected design what resource would make it executable (a stock, a reader, an environment).
4. **Information of controls.** Extend the surrogate and the information-gain model with the confounds the control arms exclude (agent effects, vehicle effects, plate position), so that their value is quantified.
5. **Import the upper ontology.** Replace the BFO-style upper level by BFO and OBI imports, and align the design classes with OBI's study-design terms.
6. **Broaden the taxonomy.** Add blinding, blocking, replicate kinds, dose–response and time-course designs, and the in vivo and clinical branches (arms × epochs).
7. **Shapes.** Express the closed-world checks as SHACL shapes over the reasoned ABox, so that they are portable beyond the RSA.
8. **Reproducibility.** Add the generators to `rsa.reproduce` and record the ontology, ABox and reasoner results in the manifest.

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A The TBox in DL notation

The complete TBox as generated from `rsa/experiment_ontology.py`. Concepts are set in sans, roles in *italics*.

Upper level

A BFO-style upper level. Continuants (material entities, qualities, realizable entities, information content entities) and occurrents (processes) are disjoint; planned processes are distinct from biological processes.

MaterialEntity ⊑ Continuant

SpecificallyDependentContinuant ⊑ Continuant

Quality ⊑ SpecificallyDependentContinuant

RealizableEntity ⊑ SpecificallyDependentContinuant

InformationContentEntity ⊑ Continuant

Process ⊑ Occurrent

Continuant ⊓ Occurrent ⊑ ⊥

Disjoint(MaterialEntity, SpecificallyDependentContinuant, InformationContentEntity)

Quality ⊓ RealizableEntity ⊑ ⊥

Role ⊑ RealizableEntity

Disposition ⊑ RealizableEntity
Role ⊓ Disposition ⊑ ⊥
PlannedProcess ⊑ Process
BiologicalProcess ⊑ Process
PlannedProcess ⊓ BiologicalProcess ⊑ ⊥
PlanSpecification ⊑ InformationContentEntity
DataItem ⊑ InformationContentEntity
Hypothesis ⊑ InformationContentEntity
Objective ⊑ InformationContentEntity
Conclusion ⊑ InformationContentEntity
Disjoint(PlanSpecification, DataItem, Hypothesis, Objective, Conclusion)

Relations and partonomy

Parthood is transitive and kind-preserving: processes have processes as parts, material entities material entities, information entities information entities. Its kinds are simple sub-roles (so they may be counted). Location, membership, participation, execution and bearing are not parthood.

Trans(*hasPart*)
partOf ≡ *hasPart*⁻
hasPhase ⊑ *hasPart*
hasStep ⊑ *hasPart*
hasWell ⊑ *hasPart*
hasOrganoid ⊑ *hasPart*
hasCell ⊑ *hasPart*
hasMedium ⊑ *hasPart*
hasArm ⊑ *hasPart*
hasComparisonSpec ⊑ *hasPart*
hasAdministrationSpec ⊑ *hasPart*
hasObjective ⊑ *hasPart*
Process ⊑ ∀*hasPart*.Process
MaterialEntity ⊑ ∀*hasPart*.MaterialEntity
InformationContentEntity ⊑ ∀*hasPart*.InformationContentEntity
containedIn ≡ *contains*⁻
containedIn ∘ *partOf* ⊑ *locatedIn*
participatesIn ≡ *hasParticipant*⁻
hasPart ∘ *hasParticipant* ⊑ *hasParticipant*
hasSpecifiedInput ⊑ *hasParticipant*
inheresIn ≡ *bearerOf*⁻
Func(*inheresIn*)
hasAssignedUnit ≡ *assignedTo*⁻
Func(*assignedTo*)
Func(*usesAgent*)
Trans(*precedes*)
∃*executes*.⊤ ⊑ PlannedProcess
⊤ ⊑ ∀*executes*.PlanSpecification
∃*isAbout*.⊤ ⊑ InformationContentEntity
∃*bearerOf*.⊤ ⊑ Continuant
⊤ ⊑ ∀*bearerOf*.SpecificallyDependentContinuant
∃*realizedIn*.⊤ ⊑ RealizableEntity
⊤ ⊑ ∀*realizedIn*.Process
Func($dose_a\,u$)
Func($atTime_h$)
Func($fromTime_h$)
Func($toTime_h$)
Func($expectedInformationGain_b\,its$)
Func($radius_u\,m$)
Func(*wellRow*)
Func(*wellColumn*)

Biological material and labware

An organoid culture is organoids in medium; an organoid has cells. A well holds at most one culture; a culture is in a well, not part of it. Chemical agents bear roles in the experiment (challenge, rescue, vehicle) and dispositions (toxicity); size is a quality of organoids, viability and recovery capacity are dispositions of cultures.

OrganoidCulture ⊑ MaterialEntity ⊓ ∃*hasOrganoid*.Organoid ⊓ ∃*hasMedium*.CultureMedium
Organoid ⊑ MaterialEntity ⊓ ∃*hasCell*.Cell
Cell ⊑ MaterialEntity
CultureMedium ⊑ MaterialEntity
ChemicalAgent ⊑ MaterialEntity
Plate ⊑ MaterialEntity ⊓ ∃*hasWell*.Well
Well ⊑ MaterialEntity ⊓ (≤ 1*contains*.OrganoidCulture)
Disjoint(OrganoidCulture, Organoid, Cell, CultureMedium, ChemicalAgent, Plate, Well)
OrganoidSize ⊑ Quality ⊓ ∃*inheresIn*.Organoid
Viability ⊑ Disposition ⊓ ∃*inheresIn*.OrganoidCulture ⊓ ∀*realizedIn*.CultureGrowth
RecoveryCapacity ⊑ Disposition ⊓ ∃*inheresIn*.OrganoidCulture ⊓ ∀*realizedIn*.RecoveryProcess
ToxicityDisposition ⊑ Disposition ⊓ ∃*inheresIn*.ChemicalAgent ⊓ ∀*realizedIn*.InjuryProcess
Toxicant ≡ ChemicalAgent ⊓ ∃*bearerOf*.ToxicityDisposition
ChallengeAgentRole ⊑ Role

RescueAgentRole ⊑ Role

VehicleRole ⊑ Role

ExperimentalUnitRole ⊑ Role

Disjoint(ChallengeAgentRole, RescueAgentRole, VehicleRole, ExperimentalUnitRole)

ChallengeAgent ≡ ChemicalAgent ⊓ ∃*bearerOf*.ChallengeAgentRole

RescueAgent ≡ ChemicalAgent ⊓ ∃*bearerOf*.RescueAgentRole

Vehicle ≡ ChemicalAgent ⊓ ∃*bearerOf*.VehicleRole

Disjoint(ChallengeAgent, RescueAgent, Vehicle)

ChallengeAgent ⊑ Toxicant

ExperimentalUnit ≡ OrganoidCulture ⊓ ∃*bearerOf*.ExperimentalUnitRole

Biological processes

Processes in which the cultures participate. A loss of recovery capacity (a point of no return) is a process type that hypotheses may posit; nothing here asserts that it occurs.

InjuryProcess ⊑ BiologicalProcess ⊓ ∃*hasParticipant*.OrganoidCulture

ProtectiveResponse ⊑ BiologicalProcess ⊓ ∃*hasParticipant*.OrganoidCulture

RecoveryProcess ⊑ BiologicalProcess ⊓ ∃*hasParticipant*.OrganoidCulture

CellDeath ⊑ BiologicalProcess ⊓ ∃*hasParticipant*.OrganoidCulture

CultureGrowth ⊑ BiologicalProcess ⊓ ∃*hasParticipant*.OrganoidCulture

LossOfRecoveryCapacity ⊑ BiologicalProcess ⊓ ∃*hasParticipant*.OrganoidCulture

Disjoint(InjuryProcess, ProtectiveResponse, RecoveryProcess, CellDeath, CultureGrowth)

Plans: designs, arms, administrations, comparisons

A study design has arms and comparisons; an arm specifies the administrations its units receive (agent, dose, time). Arm types are defined by what is administered, comparison types by the arms compared, design types by their arms and comparisons; the reasoner classifies them. Replication and the information target are ILLUSTRATIVE design criteria.

StudyDesign ⊑ PlanSpecification ⊓ ∃*hasObjective*.Objective ⊓ ∃*hasArm*.ArmSpecification

ArmSpecification ⊑ PlanSpecification ⊓ ∀*hasAdministrationSpec*.AdministrationSpecification

AdministrationSpecification ⊑ PlanSpecification ⊓ (= 1 *usesAgent*.ChemicalAgent) ⊓ ∃$dose_a u$.decimal ⊓ ∃$atTime_h$.decimal

ComparisonSpecification ⊑ PlanSpecification ⊓ (= 2 *compares*.ArmSpecification)

Disjoint(StudyDesign, ArmSpecification, AdministrationSpecification, ComparisonSpecification)

ChallengeSpec ≡ AdministrationSpecification ⊓ ∃*usesAgent*.ChallengeAgent

RescueSpec ≡ AdministrationSpecification ⊓ ∃*usesAgent*.RescueAgent

VehicleSpec ≡ AdministrationSpecification ⊓ ∃*usesAgent*.Vehicle

UntreatedArm ≡ ArmSpecification ⊓ ¬∃*hasAdministrationSpec*.AdministrationSpecification

VehicleControlArm ≡ ArmSpecification ⊓ ∃*hasAdministrationSpec*.VehicleSpec ⊓ ¬∃*hasAdministrationSpec*.(ChallengeSpec ⊔ RescueSpec)

NegativeControlArm ≡ UntreatedArm ⊔ VehicleControlArm

ChallengeOnlyArm ≡ ArmSpecification ⊓ ∃*hasAdministrationSpec*.ChallengeSpec ⊓ ¬∃*hasAdministrationSpec*.RescueSpec

RescueOnlyArm ≡ ArmSpecification ⊓ ∃*hasAdministrationSpec*.RescueSpec ⊓ ¬∃*hasAdministrationSpec*.ChallengeSpec

ChallengeRescueArm ≡ ArmSpecification ⊓ ∃*hasAdministrationSpec*.ChallengeSpec ⊓ ∃*hasAdministrationSpec*.RescueSpec

ReplicatedArm ≡ ArmSpecification ⊓ (≥ 3 *hasAssignedUnit*.ExperimentalUnit)

InjuryValidityComparison ≡ ComparisonSpecification ⊓ ∃*compares*.ChallengeOnlyArm ⊓ ∃*compares*.NegativeControlArm

RescueEffectComparison ≡ ComparisonSpecification ⊓ ∃*compares*.ChallengeRescueArm ⊓ ∃*compares*.ChallengeOnlyArm

AgentAloneComparison ≡ ComparisonSpecification ⊓ ∃*compares*.RescueOnlyArm ⊓ ∃*compares*.NegativeControlArm

RescueTimingComparison ≡ ComparisonSpecification ⊓ (= 2 *compares*.ChallengeRescueArm)

DiscriminationObjective ≡ Objective ⊓ (≥ 2 *isAbout*.Hypothesis)

Factors, levels and treatments

A factor is what the experimenter varies (here: the challenge, the time of the rescue); its levels are the values it takes, each specifying administrations. An arm's treatment is one level of every factor (atLevel; a level is shared by arms, so this is not parthood). A comparison contrasts the factors in which its two arms differ; a well-formed comparison contrasts exactly one. Which factors a comparison contrasts, and whether an arm's schedule is the union of its levels, are computed under the closed world and asserted.

Factor $\sqsubseteq$ PlanSpecification $\sqcap (\geq 2\, hasLevel.\text{FactorLevel})$

FactorLevel $\sqsubseteq$ PlanSpecification $\sqcap (= 1\, levelOf.\text{Factor})$

$levelOf \equiv hasLevel^{-}$

$hasLevel \sqsubseteq hasPart$

$hasFactor \sqsubseteq hasPart$

ArmSpecification $\sqsubseteq \forall atLevel.\text{FactorLevel}$

ComparisonSpecification $\sqsubseteq \forall contrasts.\text{Factor}$

SingleFactorComparison $\equiv$ ComparisonSpecification $\sqcap (= 1\, contrasts.\text{Factor})$

FactorialDesign $\equiv$ StudyDesign $\sqcap (\geq 2\, hasFactor.\text{Factor})$

WellContrastedDesign $\equiv$ StudyDesign $\sqcap \forall hasComparisonSpec.\text{SingleFactorComparison}$

InteractionContrast $\sqsubseteq$ PlanSpecification $\sqcap (= 2\, differenceOf.\text{ComparisonSpecification})$

$hasInteractionContrast \sqsubseteq hasPart$

InteractionAwareDesign $\equiv$ StudyDesign $\sqcap \exists hasInteractionContrast.\text{InteractionContrast}$

Disjoint(Factor, FactorLevel, StudyDesign, ArmSpecification, AdministrationSpecification, ComparisonSpecification, InteractionContrast)

Controls

A control is defined by what it controls for. Negative controls (untreated, vehicle) give the baseline; the agent-alone arm separates the rescue agent's own effect from rescue of injury; a positive control is an arm whose outcome is expected from prior evidence, so that a null result elsewhere can be interpreted (the agent and the assay worked in this run).

Expectation $\sqsubseteq$ DataItem

PriorEvidence $\sqsubseteq$ DataItem

EvidenceBasedExpectation $\equiv$ Expectation $\sqcap \exists derivedFrom.\text{PriorEvidence}$

PositiveControlArm $\equiv$ ArmSpecification $\sqcap \exists hasExpectedOutcome.\text{EvidenceBasedExpectation}$

PositiveControlComparison $\equiv$ ComparisonSpecification $\sqcap \exists compares.\text{PositiveControlArm} \sqcap \exists compares.\text{ChallengeOnlyArm}$

PositiveControlledDesign $\equiv$ StudyDesign $\sqcap \exists hasComparisonSpec.\text{PositiveControlComparison}$

Readouts and analysis plan

A readout (endpoint) specifies what is measured, when and by which assay, and what it is a proxy for. The analysis plan, part of the design and fixed before the run, has one decision rule per comparison (a threshold test, a Bayesian update over the hypotheses, or an estimate) and a rule for deviations. A design is pre-specified when every comparison is covered by a rule.

$hasReadout \sqsubseteq hasPart$

$hasAnalysisPlan \sqsubseteq hasPart$

$hasDecisionRule \sqsubseteq hasPart$

ReadoutSpecification $\sqsubseteq$ PlanSpecification $\sqcap \exists measures.\text{Quality} \sqcap \exists atTime_{h}.\text{decimal}$

SizeReadout $\equiv$ ReadoutSpecification $\sqcap \exists measures.\text{OrganoidSize} \sqcap \exists isProxyFor.\text{Viability}$

AnalysisPlan $\sqsubseteq$ PlanSpecification $\sqcap \exists hasDecisionRule.\text{DecisionRule}$

DecisionRule $\sqsubseteq$ PlanSpecification

ComparisonRule $\sqsubseteq$ DecisionRule $\sqcap (= 1\, appliesTo.\text{ComparisonSpecification})$

ThresholdRule $\sqsubseteq$ ComparisonRule

EstimationRule $\sqsubseteq$ ComparisonRule

BayesianUpdateRule $\sqsubseteq$ ComparisonRule $\sqcap (\geq 2\, isAbout.\text{Hypothesis})$

DeviationRule $\sqsubseteq$ DecisionRule $\sqcap \neg \exists appliesTo.\top$

Disjoint(ThresholdRule, EstimationRule, BayesianUpdateRule, DeviationRule)

Disjoint(ReadoutSpecification, AnalysisPlan, DecisionRule, Factor, StudyDesign, ArmSpecification, ComparisonSpecification)

CoveredComparison ≡ ComparisonSpecification ⊓ ∃*appliesTo*⁻.ComparisonRule

PreSpecifiedDesign ≡ StudyDesign ⊓ ∃*hasReadout*.ReadoutSpecification ⊓ ∃*hasAnalysisPlan*.AnalysisPlan ⊓ ∀*hasComparisonSpec*.CoveredComparison

DeviationAwareDesign ≡ PreSpecifiedDesign ⊓ ∃*hasAnalysisPlan*.∃*hasDecisionRule*.DeviationRule

Design classes

Design types are defined by their arms, comparisons, factors and analysis plan; the reasoner places every design under each class it satisfies. Adequate control (the reviewer's standard) requires the agent-alone and vehicle controls, a positive control, replication, single-factor comparisons and a pre-specified analysis. Replication and the information target are ILLUSTRATIVE criteria.

ComparativeDesign ≡ StudyDesign ⊓ (≥ 2*hasArm*.ArmSpecification) ⊓ ∃*hasComparisonSpec*.ComparisonSpecification

ControlledDesign ≡ ComparativeDesign ⊓ ∃*hasArm*.NegativeControlArm

InjuryRescueDesign ≡ ControlledDesign ⊓ ∃*hasArm*.ChallengeRescueArm ⊓ ∃*hasArm*.ChallengeOnlyArm ⊓ ∃*hasComparisonSpec*.RescueEffectComparison

ValidatedInjuryRescueDesign ≡ InjuryRescueDesign ⊓ ∃*hasComparisonSpec*.InjuryValidityComparison

AgentControlledInjuryRescueDesign ≡ ValidatedInjuryRescueDesign ⊓ ∃*hasArm*.RescueOnlyArm ⊓ ∃*hasArm*.VehicleControlArm ⊓ ∃*hasComparisonSpec*.AgentAloneComparison

TherapeuticWindowDesign ≡ InjuryRescueDesign ⊓ (≥ 2*hasArm*.ChallengeRescueArm)

ReplicatedDesign ≡ StudyDesign ⊓ ∀*hasArm*.ReplicatedArm

AdequatelyControlledInjuryRescueDesign ≡ AgentControlledInjuryRescueDesign ⊓ PositiveControlledDesign ⊓ ReplicatedDesign ⊓ WellContrastedDesign ⊓ PreSpecifiedDesign

HypothesisDiscriminatingDesign ≡ StudyDesign ⊓ ∃*hasObjective*.DiscriminationObjective

SufficientlyInformativeDesign ≡ StudyDesign ⊓ ∃*expectedInformationGain*$_b$*its*.decimal$_{\geq 0.5}$

Experiments, phases and steps

An experiment (investigation) is a planned process that executes exactly one study design; its classification follows its design's. Its parts are three phases, their parts the steps: administrations and incubations, assays, data transformations.

Investigation ⊑ PlannedProcess ⊓ (= 1*executes*.StudyDesign)

ComparativeExperiment ≡ Investigation ⊓ ∃*executes*.ComparativeDesign

ControlledExperiment ≡ Investigation ⊓ ∃*executes*.ControlledDesign

InjuryRescueExperiment ≡ Investigation ⊓ ∃*executes*.InjuryRescueDesign

ValidatedInjuryRescueExperiment ≡ Investigation ⊓ ∃*executes*.ValidatedInjuryRescueDesign

AdequatelyControlledInjuryRescueExperiment ≡ Investigation ⊓ ∃*executes*.AdequatelyControlledInjuryRescueDesign

TherapeuticWindowExperiment ≡ Investigation ⊓ ∃*executes*.TherapeuticWindowDesign

FactorialExperiment ≡ Investigation ⊓ ∃*executes*.FactorialDesign

HypothesisDiscriminatingExperiment ≡ Investigation ⊓ ∃*executes*.HypothesisDiscriminatingDesign

ComparativeExperiment ⊑ (= 1*hasPhase*.CultivationPhase) ⊓ (= 1*hasPhase*.ObservationPhase) ⊓ (= 1*hasPhase*.AnalysisPhase)

Phase ⊑ PlannedProcess

CultivationPhase ⊑ Phase ⊓ ∀*hasStep*.(Administration ⊔ Incubation)

ObservationPhase ⊑ Phase ⊓ ∀*hasStep*.Assay

AnalysisPhase ⊑ Phase ⊓ ∀*hasStep*.DataTransformation

Disjoint(CultivationPhase, ObservationPhase, AnalysisPhase)

Administration ⊑ PlannedProcess ⊓ (= 1*executes*.AdministrationSpecification) ⊓ ∃*hasSpecifiedInput*.ChemicalAgent ⊓ ∃*hasSpecifiedInput*.ExperimentalUnit

ChallengeAdministration ≡ Administration ⊓ ∃*executes*.ChallengeSpec

RescueAdministration ≡ Administration ⊓ ∃*executes*.RescueSpec

Incubation ⊑ PlannedProcess ⊓ ∃*hasParticipant*.OrganoidCulture

Assay ⊑ PlannedProcess ⊓ ∃*hasSpecifiedInput*.ExperimentalUnit ⊓ ∃*hasSpecifiedOutput*.DataItem

ImagingAssay ⊑ Assay ⊓ ∃*hasSpecifiedOutput*.Image

DataTransformation ⊑ PlannedProcess

ImageAnalysis ⊑ DataTransformation ⊓ ∃*hasSpecifiedInput*.Image ⊓ ∃*hasSpecifiedOutput*.SizeDatum

StatisticalComparison ⊑ DataTransformation ⊓ (= 1*executes*.ComparisonSpecification) ⊓ ∃*executes*.ComparisonRule ⊓ ∃*hasSpecifiedOutput*.EffectEstimate

ModelUpdate ⊑ DataTransformation ⊓ ∃*hasSpecifiedOutput*.PosteriorDistribution

Disjoint(Phase, Administration, Incubation, Assay, DataTransformation)

Data, hypotheses and conclusions

Images are about units; a size datum is about organoid size and is a proxy for viability. Recovery hypotheses are about recovery processes: the graded one posits no loss of recovery capacity, the threshold one posits it, the protective one posits a protective response that delays it. Predictions derive from hypotheses; conclusions (the insights) derive from effect estimates and posteriors.

Image $\sqsubseteq$ DataItem $\sqcap \exists isAbout$.ExperimentalUnit

MeasurementDatum $\sqsubseteq$ DataItem

SizeDatum $\sqsubseteq$ MeasurementDatum $\sqcap \exists isAbout$.OrganoidSize $\sqcap \exists isProxyFor$.Viability

EffectEstimate $\sqsubseteq$ DataItem $\sqcap \exists derivedFrom$.SizeDatum

PosteriorDistribution $\sqsubseteq$ DataItem $\sqcap (\geq 2\, isAbout$.Hypothesis)

Prediction $\sqsubseteq$ Expectation $\sqcap \exists derivedFrom$.Hypothesis $\sqcap \exists isAbout$.ComparisonSpecification

Disjoint(Image, MeasurementDatum, EffectEstimate, PosteriorDistribution, Expectation, PriorEvidence)

RecoveryHypothesis $\sqsubseteq$ Hypothesis $\sqcap \exists isAbout$.RecoveryProcess

GradedRecoveryHypothesis $\sqsubseteq$ RecoveryHypothesis $\sqcap \neg\exists posits$.LossOfRecoveryCapacity

ThresholdRecoveryHypothesis $\sqsubseteq$ RecoveryHypothesis $\sqcap \exists posits$.LossOfRecoveryCapacity

ProtectiveResponseHypothesis $\sqsubseteq$ ThresholdRecoveryHypothesis $\sqcap \exists posits$.(ProtectiveResponse $\sqcap$ $\exists delays$.LossOfRecoveryCapacity)

Conclusion $\sqsubseteq \exists derivedFrom$.(EffectEstimate $\sqcup$ PosteriorDistribution)

Laboratories, devices and capabilities

A laboratory has devices as parts; a device bears functions, dispositions realized only in processes of one kind (a dosing function only in administrations, an imaging function only in imaging assays). A laboratory's capability classes are defined by the functions its devices bear, by the environments it provides and by its validated readout bindings, so the reasoner classifies laboratories from their devices. Reagent stocks are located in a laboratory, not parts of it.

Laboratory $\sqsubseteq$ MaterialEntity $\sqcap \exists hasDevice$.Device

$hasDevice \sqsubseteq hasPart$

Device $\sqsubseteq$ MaterialEntity

Disjoint(Laboratory, Device, OrganoidCulture, ChemicalAgent, Plate, Well)

DeviceFunction $\sqsubseteq$ Disposition $\sqcap \exists inheresIn$.Device

DosingFunction $\sqsubseteq$ DeviceFunction $\sqcap \forall realizedIn$.Administration

IncubationFunction $\sqsubseteq$ DeviceFunction $\sqcap \forall realizedIn$.Incubation

BrightfieldImagingFunction $\sqsubseteq$ DeviceFunction $\sqcap \forall realizedIn$.ImagingAssay

LuminescenceReadingFunction $\sqsubseteq$ DeviceFunction $\sqcap \forall realizedIn$.LuminescenceAssay

AbsorbanceReadingFunction $\sqsubseteq$ DeviceFunction $\sqcap \forall realizedIn$.AbsorbanceAssay

CalibrationFunction $\sqsubseteq$ DeviceFunction

QuarantineFunction $\sqsubseteq$ DeviceFunction

TransportFunction $\sqsubseteq$ DeviceFunction

PlateHandlingFunction $\sqsubseteq$ DeviceFunction

StorageFunction $\sqsubseteq$ DeviceFunction

EndpointAssay $\sqsubseteq$ Assay

LuminescenceAssay $\sqsubseteq$ EndpointAssay

AbsorbanceAssay $\sqsubseteq$ EndpointAssay

ImagingAssay $\sqcap$ EndpointAssay $\sqsubseteq \bot$

MetabolicActivity $\sqsubseteq$ Quality $\sqcap \exists inheresIn$.OrganoidCulture

EnvironmentProfile $\sqsubseteq$ PlanSpecification

ReadoutBinding $\sqsubseteq$ InformationContentEntity $\sqcap \exists isAbout$.Quality

ValidatedReadoutBinding $\sqsubseteq$ ReadoutBinding

DosingLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf$.DosingFunction

CultivationLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf$.IncubationFunction

BrightfieldImagingLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf$.BrightfieldImagingFunction

LuminescenceReadoutLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf$.LuminescenceReadingFunction

AbsorbanceReadoutLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf$.AbsorbanceReadingFunction

QualityControlledLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf$.CalibrationFunction

ContainmentCapableLab ≡ Laboratory ⊓ ∃*hasDevice*.∃*bearerOf*.QuarantineFunction

ConnectedLab ≡ Laboratory ⊓ ∃*hasDevice*.∃*bearerOf*.TransportFunction

MultiEnvironmentLab ≡ Laboratory ⊓ (≥ 2*providesEnvironment*.EnvironmentProfile)

ValidatedSizeReadoutLab ≡ Laboratory ⊓ ∃*hasReadoutBinding*.(ValidatedReadoutBinding ⊓ ∃*isAbout*.OrganoidSize)

ValidatedMetabolicReadoutLab ≡ Laboratory ⊓ ∃*hasReadoutBinding*.(ValidatedReadoutBinding ⊓ ∃*isAbout*.MetabolicActivity)

TreatmentAndImagingModule ≡ DosingLab ⊓ CultivationLab ⊓ BrightfieldImagingLab ⊓ ValidatedSizeReadoutLab

Design requirements

What a design needs from a laboratory follows from its content: administrations need dosing, an imaging readout needs imaging, an endpoint readout needs the corresponding reader and comes after every non-destructive observation, an environment factor needs as many controlled environments as it has levels. Requirement classes are defined, so the reasoner classifies designs by their needs. Each requirement is met by one laboratory capability class; whether the laboratories of a site meet all of a design's requirements, with its reagents, readout bindings and timing, is a closed-world question over the individuals.

RequiresDosing ≡ StudyDesign ⊓ ∃*hasArm*.∃*hasAdministrationSpec*.AdministrationSpecification

RequiresCultivation ≡ StudyDesign ⊓ ∃*hasArm*.ArmSpecification

ImagingReadout ⊑ ReadoutSpecification

EndpointReadout ⊑ ReadoutSpecification

LuminescenceReadout ⊑ EndpointReadout

AbsorbanceReadout ⊑ EndpointReadout

ImagingReadout ⊓ EndpointReadout ⊑ ⊥

RequiresImaging ≡ StudyDesign ⊓ ∃*hasReadout*.ImagingReadout

RequiresLuminescence ≡ StudyDesign ⊓ ∃*hasReadout*.LuminescenceReadout

RequiresAbsorbance ≡ StudyDesign ⊓ ∃*hasReadout*.AbsorbanceReadout

RequiresEndpointHandling ≡ StudyDesign ⊓ ∃*hasReadout*.EndpointReadout

EnvironmentFactor ⊑ Factor

RequiresMultipleEnvironments ≡ StudyDesign ⊓ ∃*hasFactor*.EnvironmentFactor

Alignment to Cell 01's biology ontology

Organoid ⊑ bio:Organoid

CultureMedium ⊑ bio:CultureMedium

Toxicant ⊑ bio:Toxicant

RescueAgent ⊑ bio:RescueAgent

ChallengeAdministration ⊑ bio:ControlledToxicChallenge

RescueAdministration ⊑ bio:RescueIntervention

ImagingAssay ⊑ bio:ImagingProcess

Incubation ⊑ bio:IncubationProcess

Hypothesis ⊑ bio:Hypothesis

Prediction ⊑ bio:Prediction

ProtectiveResponse ⊑ bio:ProtectiveResponse

RecoveryProcess ⊑ bio:RecoveryProcess

CellDeath ⊑ bio:CellDeathProcess

CultureGrowth ⊑ bio:CultureGrowth

InjuryProcess ⊑ bio:DamageAccumulation

Plate ⊑ bio:CulturePlate

Well ⊑ bio:CultureWell

B ABox of the current experiment (excerpt)

The first assertions of the current experiment's ABox, as rendered from the same source as the OWL file.

StudyDesign(`challenge_rescue_design`)
Objective(`objective`)
GradedRecoveryHypothesis(`M1_graded`)
ThresholdRecoveryHypothesis(`M2_threshold`)
ProtectiveResponseHypothesis(`M3_protective`)
ChemicalAgent(`agent_challenge`)
ChallengeAgentRole(`role_challenge`)
ChemicalAgent(`agent_rescue`)
RescueAgentRole(`role_rescue`)
ToxicityDisposition(`toxicity`)
Factor(`factor_challenge`)
FactorLevel(`level_challenge_none`)
FactorLevel(`level_challenge_0p85au_0h`)
Factor(`factor_rescue`)
FactorLevel(`level_rescue_none`)
FactorLevel(`level_rescue_0p7au_48h`)
ArmSpecification(`arm_rescued`)
AdministrationSpecification(`admin_rescued_challenge_0h`)
AdministrationSpecification(`admin_rescued_rescue_48h`)
(≤ 2*hasAdministrationSpec*.AdministrationSpecification)(`arm_rescued`)
OrganoidCulture(`unit_rescued_1`)
ExperimentalUnitRole(`unit_rescued_1_role`)
(≤ 1*hasAssignedUnit*.ExperimentalUnit)(`arm_rescued`)
ArmSpecification(`arm_comparator`)
AdministrationSpecification(`admin_comparator_challenge_0h`)
(≤ 1*hasAdministrationSpec*.AdministrationSpecification)(`arm_comparator`)

OrganoidCulture(unit_comparator_1)
ExperimentalUnitRole(unit_comparator_1_role)
(≤ 1 *hasAssignedUnit*.ExperimentalUnit)(arm_comparator)
ArmSpecification(arm_control)
(≤ 0 *hasAdministrationSpec*.AdministrationSpecification)(arm_control)
OrganoidCulture(unit_control_1)
ExperimentalUnitRole(unit_control_1_role)
(≤ 1 *hasAssignedUnit*.ExperimentalUnit)(arm_control)
(≤ 3 *hasArm*.⊤)(challenge_rescue_design)
ComparisonSpecification(cmp_validity)
(≤ 1 *contrasts*.⊤)(cmp_validity)
ComparisonSpecification(cmp_primary)
(≤ 1 *contrasts*.⊤)(cmp_primary)
(≤ 2 *hasComparisonSpec*.⊤)(challenge_rescue_design)
Viability(viability)
ReadoutSpecification(readout)
ImagingReadout(readout)
OrganoidSize(organoid_size)
AnalysisPlan(analysis_plan)
ThresholdRule(rule_validity)
BayesianUpdateRule(rule_primary)
Plate(plate_P17)
Well(well_A01)
Well(well_A02)
Well(well_A05)
Investigation(experiment_challenge_rescue)
CultivationPhase(phase_cultivation)
ObservationPhase(phase_observation)
AnalysisPhase(phase_analysis)
Administration(step_x1)
Administration(step_x2)
Incubation(step_g1)
Administration(step_x3)
Incubation(step_g2)
ImagingAssay(step_o1)
Image(image_rescued)
ImagingAssay(step_o2)
Image(image_comparator)
ImagingAssay(step_o3)
Image(image_control)
ImageAnalysis(step_i1)
SizeDatum(size_rescued)
ImageAnalysis(step_i2)
SizeDatum(size_comparator)
ImageAnalysis(step_i3)
SizeDatum(size_control)
StatisticalComparison(step_v1)
EffectEstimate(effect_validity)
StatisticalComparison(step_a1)
EffectEstimate(effect_primary)
ModelUpdate(step_a1)
PosteriorDistribution(posterior)
Prediction(prediction_M1_graded)
Prediction(prediction_M2_threshold)
Prediction(prediction_M3_protective)
Conclusion(conclusion_answer)
Conclusion(conclusion_rescue_effect)
Conclusion(conclusion_challenge_effect)
hasObjective(challenge_rescue_design, objective)
isAbout(objective, M1_graded)
isAbout(objective, M2_threshold)
isAbout(objective, M3_protective)
bearerOf(agent_challenge, role_challenge)
bearerOf(agent_rescue, role_rescue)
bearerOf(agent_challenge, toxicity)
hasFactor(challenge_rescue_design, factor_challenge)
hasLevel(factor_challenge, level_challenge_none)
hasLevel(factor_challenge, level_challenge_0p85au_0h)
hasFactor(challenge_rescue_design, factor_rescue)
hasLevel(factor_rescue, level_rescue_none)
hasLevel(factor_rescue, level_rescue_0p7au_48h)
hasArm(challenge_rescue_design, arm_rescued)
atLevel(arm_rescued, level_challenge_0p85au_0h)
atLevel(arm_rescued, level_rescue_0p7au_48h)
hasAdministrationSpec(arm_rescued, admin_rescued_challenge_0h)
usesAgent(admin_rescued_challenge_0h, agent_challenge)
hasAdministrationSpec(arm_rescued, admin_rescued_rescue_48h)
usesAgent(admin_rescued_rescue_48h, agent_rescue)
bearerOf(unit_rescued_1, unit_rescued_1_role)
assignedTo(unit_rescued_1, arm_rescued)
hasArm(challenge_rescue_design, arm_comparator)
atLevel(arm_comparator, level_challenge_0p85au_0h)
atLevel(arm_comparator, level_rescue_none)
hasAdministrationSpec(arm_comparator, admin_comparator_challenge_0h)
usesAgent(admin_comparator_challenge_0h, agent_challenge)
bearerOf(unit_comparator_1, unit_comparator_1_role)
assignedTo(unit_comparator_1, arm_comparator)
hasArm(challenge_rescue_design, arm_control)
atLevel(arm_control, level_challenge_none)
atLevel(arm_control, level_rescue_none)
bearerOf(unit_control_1, unit_control_1_role)
assignedTo(unit_control_1, arm_control)
hasComparisonSpec(challenge_rescue_design, cmp_validity)
compares(cmp_validity, arm_comparator)

C OWL functional syntax (excerpt)

```
SubObjectPropertyOf(ObjectPropertyChain(xp:containedIn xp:partOf) xp:locatedIn)
SubObjectPropertyOf(ObjectPropertyChain(xp:hasPart xp:hasParticipant) xp:hasParticipant)
EquivalentClasses(xp:ChallengeOnlyArm ObjectIntersectionOf(xp:ArmSpecification ObjectSomeValuesFrom(xp:hasAdministrationSpec xp:ChallengeSpec)
↪  ObjectComplementOf(ObjectSomeValuesFrom(xp:hasAdministrationSpec xp:RescueSpec))))
EquivalentClasses(xp:SingleFactorComparison ObjectIntersectionOf(xp:ComparisonSpecification ObjectExactCardinality(1 xp:contrasts xp:Factor)))
EquivalentClasses(xp:AdequatelyControlledInjuryRescueDesign ObjectIntersectionOf(xp:AgentControlledInjuryRescueDesign xp:PositiveControlledDesign xp:ReplicatedDesign
↪  xp:WellContrastedDesign xp:PreSpecifiedDesign))
EquivalentClasses(xp:AdequatelyControlledInjuryRescueExperiment ObjectIntersectionOf(xp:Investigation ObjectSomeValuesFrom(xp:executes xp:AdequatelyControlledInjuryRescueDesign)))
ClassAssertion(xp:ArmSpecification xp:arm_comparator)
ClassAssertion(ObjectMaxCardinality(1 xp:hasAdministrationSpec xp:AdministrationSpecification) xp:arm_comparator)
ClassAssertion(ObjectMaxCardinality(1 xp:hasAssignedUnit xp:ExperimentalUnit) xp:arm_comparator)
```

Listing 5: Selected axioms and closure assertions in OWL functional syntax (`rsa/out/experiment_ontology/experiment.ofn`).

D Files and rebuilding

To rebuild, run from the repository root:

- `env_cad/bin/python -m rsa.experiment_designator`
- `env_cad/bin/python -m rsa.experiment_concepts`
- `env_cad/bin/python -m rsa.experiment_ontology`
- `env_cad/bin/python -m rsa.lab_affordance`
- `env_cad/bin/python report/experiments/build_experiments_report.py`

The report builder re-runs all generators first (`--no-regen` skips them).

Table 22: Files of the experiment layer.

file	content
rsa/registry/lab/experiment.ldl	the authored experiment designator
rsa/registry/lab/experiment_concepts.ldl	LDL concept hierarchy with partonomy
rsa/experiment_designator.py	logic check, action derivation, top-level designator
rsa/experiment_concepts.py	closed-world conformance, classification, partonomy figure
rsa/experiment_dl.py	first DL statement (v1)
rsa/experiment_ontology.py	OWL 2 DL axiomatization, HermiT reasoning, closed-world checks (v2)
rsa/culture_plan.py, rsa/culture_to_cram.py	culture-level plan; CRAM/RPL controller
rsa/lab_affordance.py	laboratory capability profiles, design requirements, matching
rsa/out/experiment_ontology/	experiment.ofn, experiment.owl, experiment_owl.pdf, lab_affordance.pdf, JSON results
report/experiments/build_experiments_report.py	this report